Steroidogenesis

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Learning Objectives

1. Map the pathways involved in synthesis of adrenal steroids and explain how tissue-specific production of corticosteroids is achieved.

2. Describe the hormonal regulation of adrenal steroid production and explain why chronic signaling leads to congenital adrenal hyperplasia (CAH).

3. Map the pathways involved in the synthesis of testosterone, estrogen and progesterone in gonadal tissue and describe their hormonal regulation.

4. Compare and contrast the effects on blood pressure, metabolism, stress response, and sexual development of common disorders of steroidogenesis (21-hydroxylase deficiency, 11β-hydroxylase deficiency, 17α-hydroxylase deficiency, 11β-hydroxysteroid dehydrogenase deficiency) and describe their pathophysiological mechanisms.

5. Outline common laboratory tests for identifying patients with defects in steroidogenesis and explain how they discriminate between 21-hydroxylase deficiency, 11β-hydroxylase deficiency and 17α-hydroxylase deficiency.

What are steroids?

Steroids are cell-permeable hormones derived from cholesterol that bind to and activate nuclear hormone receptors.

Four types:

- **Mineralocorticoids** – important for electrolyte and blood pressure regulation (e.g., aldosterone)
- **Glucocorticoids** – important for the stress response, glucose handling, metabolism regulation and the inflammatory response (e.g., cortisol)
- **Androgens** – important for the development and maintenance of male characteristics (e.g., testosterone)
- **Estrogens** – important for the development and regulation of the female reproductive system and secondary sexual characteristics (e.g., estrogen)

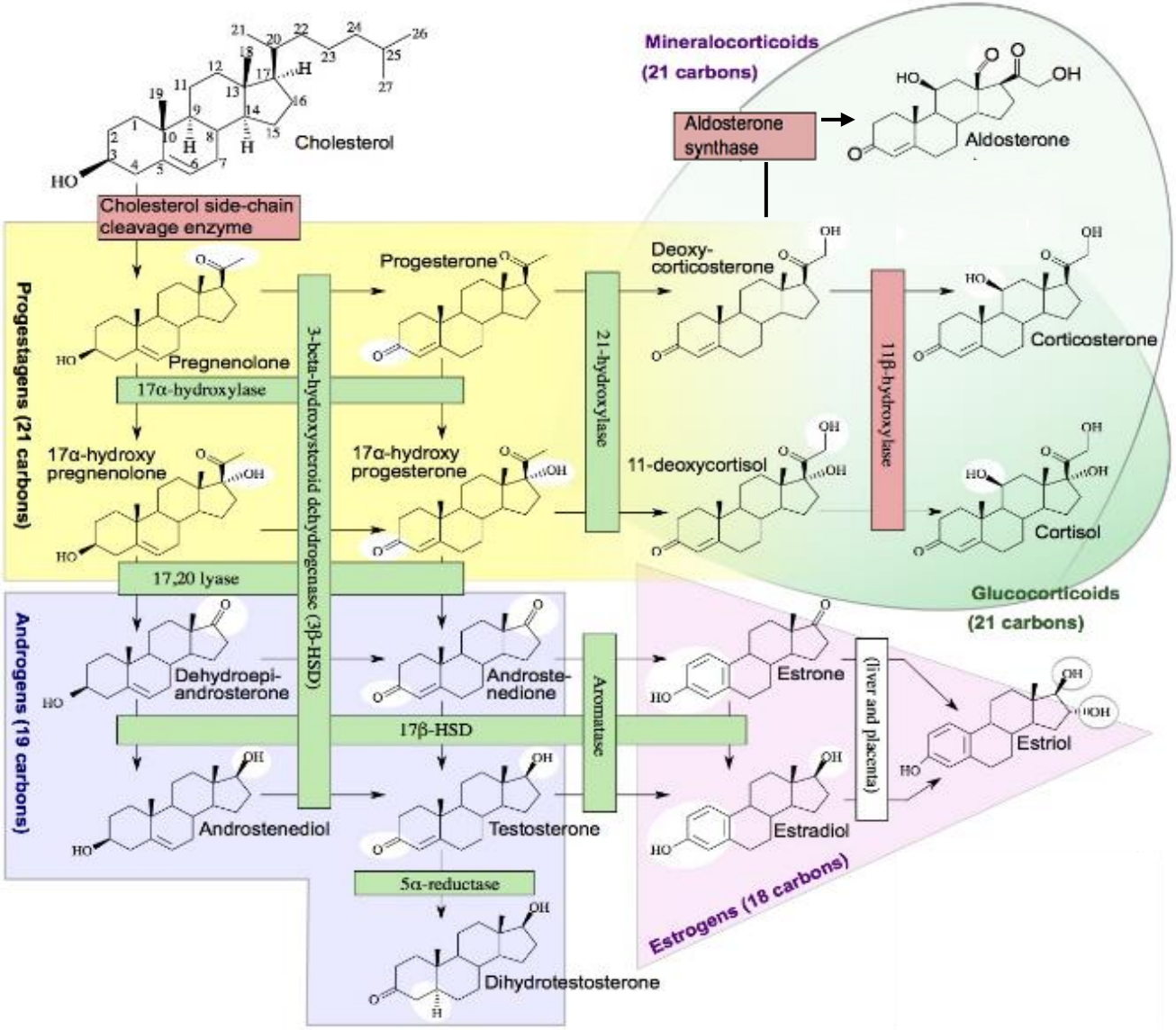
Why should you care about steroidogenesis?

Steroid hormones play significant roles in electrolyte balance, metabolism, and development, and their dysregulation can lead to severe or life-threatening pathologies.

**Questions related to clinical impact, diagnosis and treatment of genetic defects in steroidogenesis are common on Step 1.*

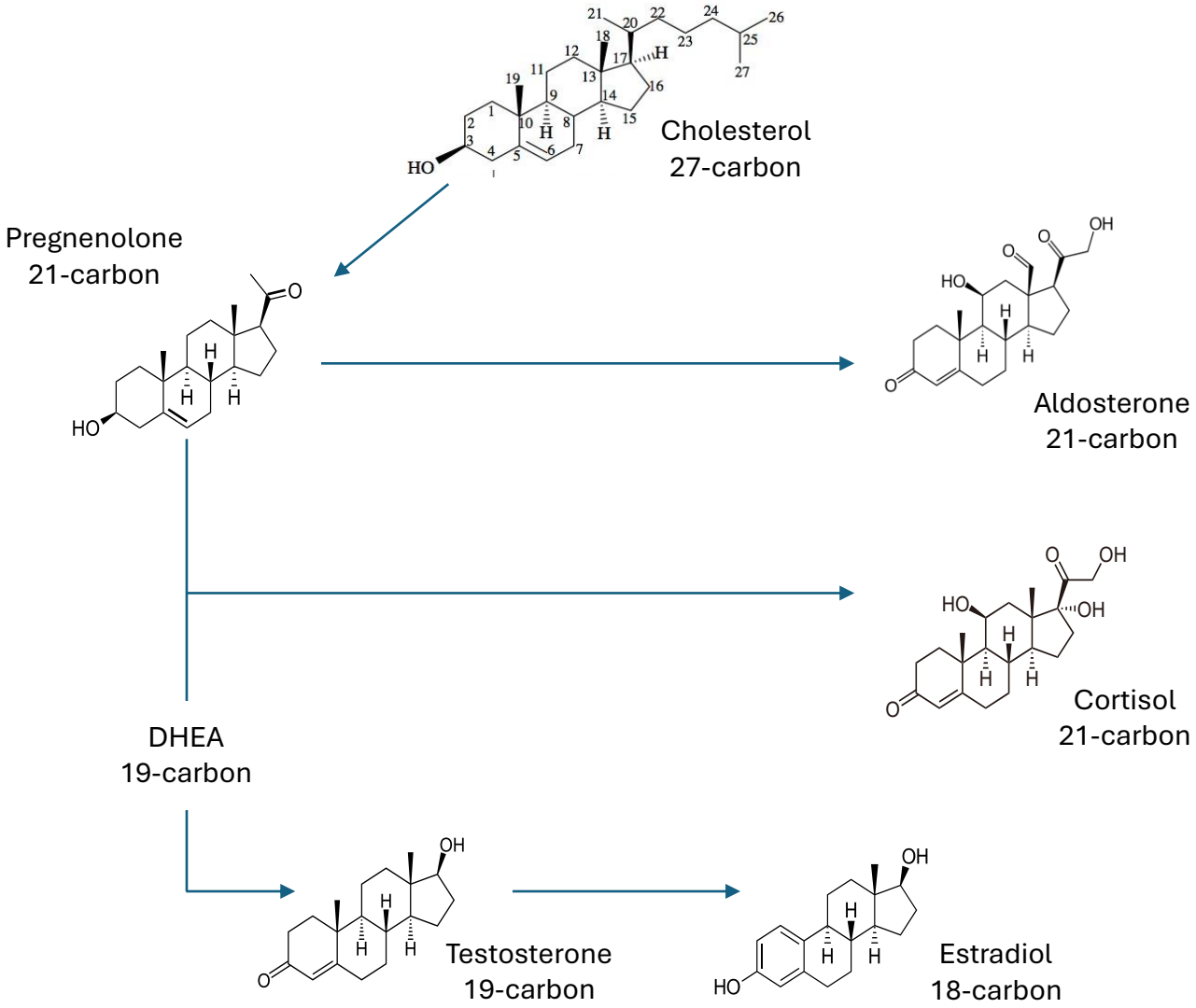
Steroidogenesis Overview

- NO cell performs *all* of these reactions.
- Expression patterns of enzymes and cofactors/enzyme modifiers determine tissue-specific steroid production.
- Intermediates can diffuse between cells to complete synthesis.



Steroidogenesis Overview

- In no cell do all of these reactions happen.
- Expression patterns of enzymes and cofactors/enzyme modifiers determine tissue-specific steroid production.
- Intermediates can diffuse between cells to complete synthesis



Case 1: A 10-day-old infant presented to the Emergency Department with severe electrolyte imbalance and life-threatening arrhythmia. The parents reported a 3-day history of poor feeding and lethargy. On examination, he was bradycardic (heart rate of 65 beats/min) with signs of dehydration. Laboratory values indicate…hyponatraemia and hyperkalaemia.

(BMJ Case Rep. 2019; 12(1): e227565; doi: 10.1136/bcr-2018-227565)

Question: What is the differential diagnosis for lethargy, electrolyte imbalance and arrhythmia in a newborn?

1) Neonatal sepsis
2) Congenital heart disease
3) Fatty acid oxidation/urea cycle defects
4) **Congenital adrenal hyperplasia**

Learning Objectives

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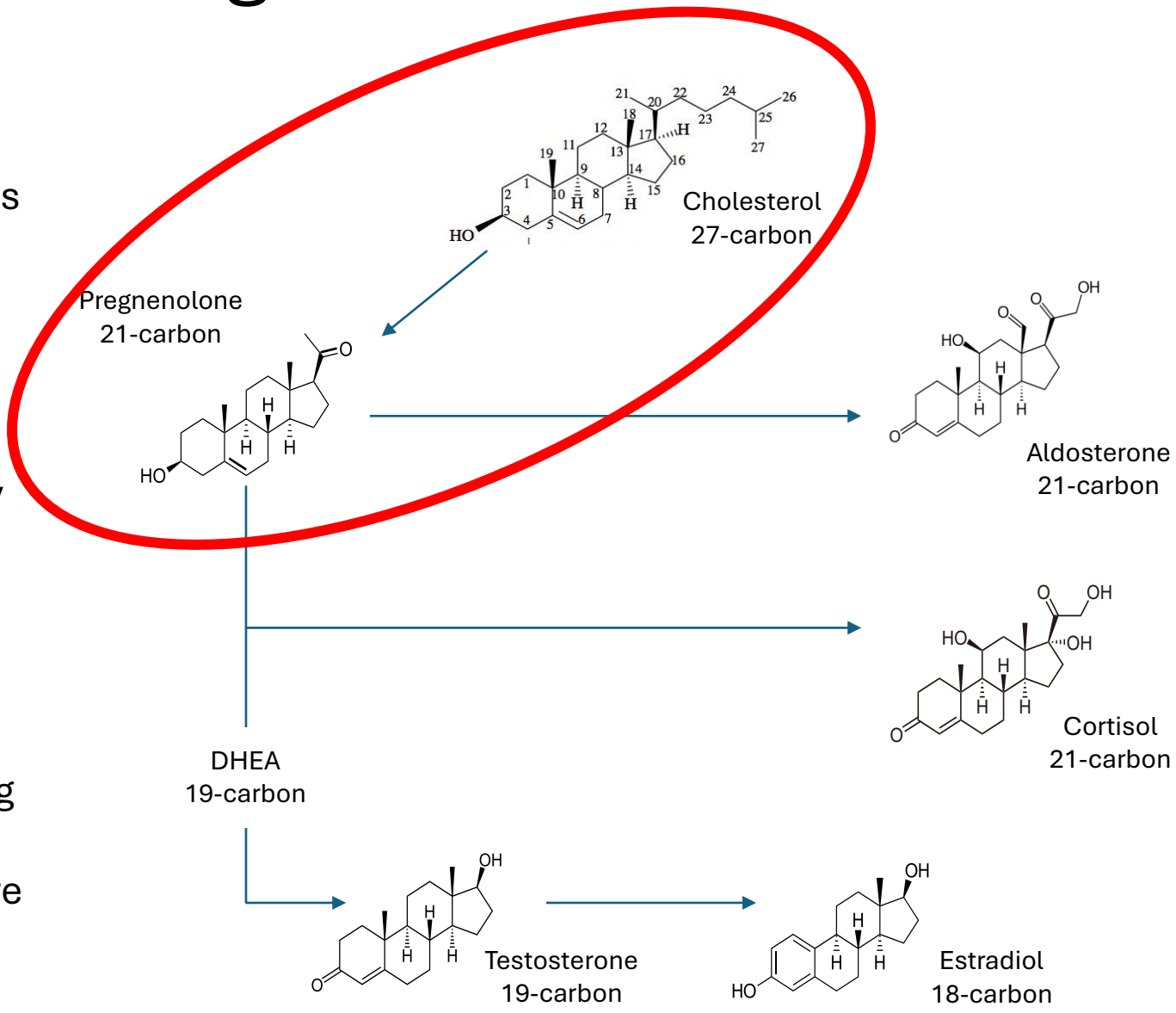
3. Map the pathways involved in the synthesis of testosterone, estrogen and progesterone in gonadal tissue and describe their hormonal regulation.

4. Compare and contrast the effects on blood pressure, metabolism, stress response, and sexual development of common disorders of steroidogenesis (21-hydroxylase deficiency, 11β-hydroxylase deficiency, 17α-hydroxylase deficiency, 11β-hydroxysteroid dehydrogenase deficiency) and describe their pathophysiological mechanisms.

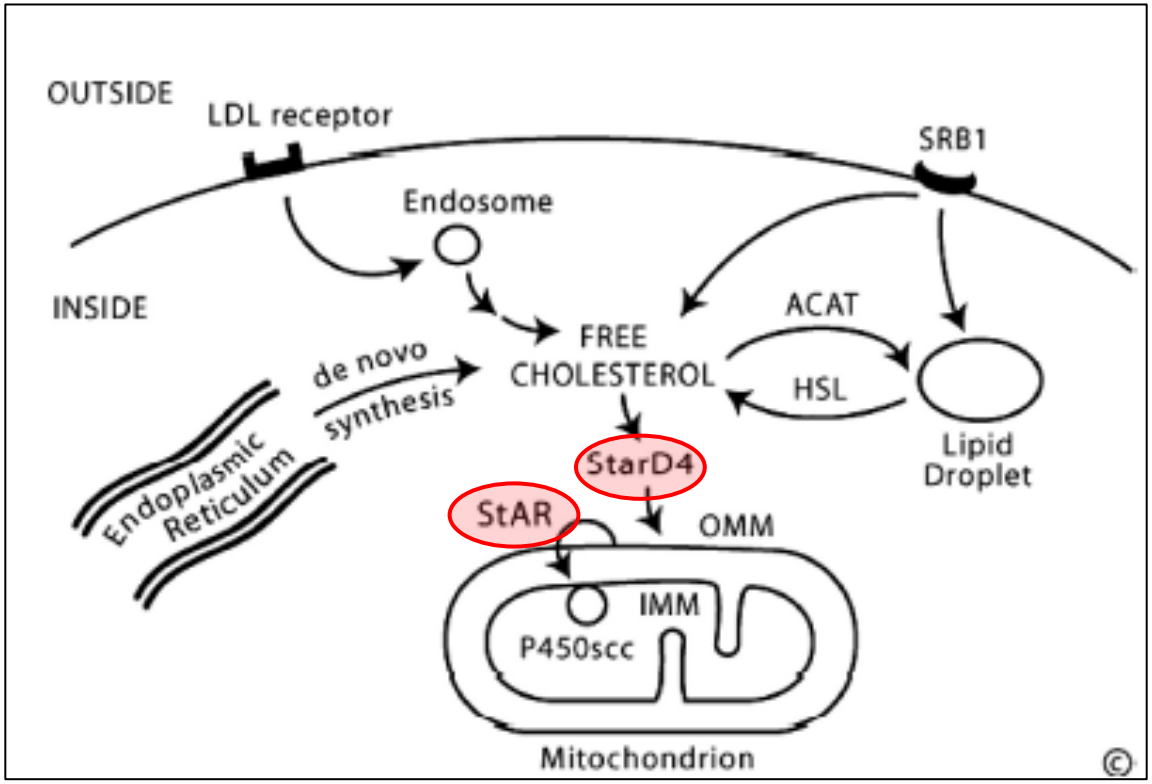
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Steroidogenesis Overview

- By definition, steroidogenic tissues can steroids *de novo* from cholesterol (distinguishing steroidogenic tissues from other tissues that merely modify steroid hormones).
- The first steps of steroidogenesis involve transporting cholesterol to the mitochondria where the acyl chain is removed.

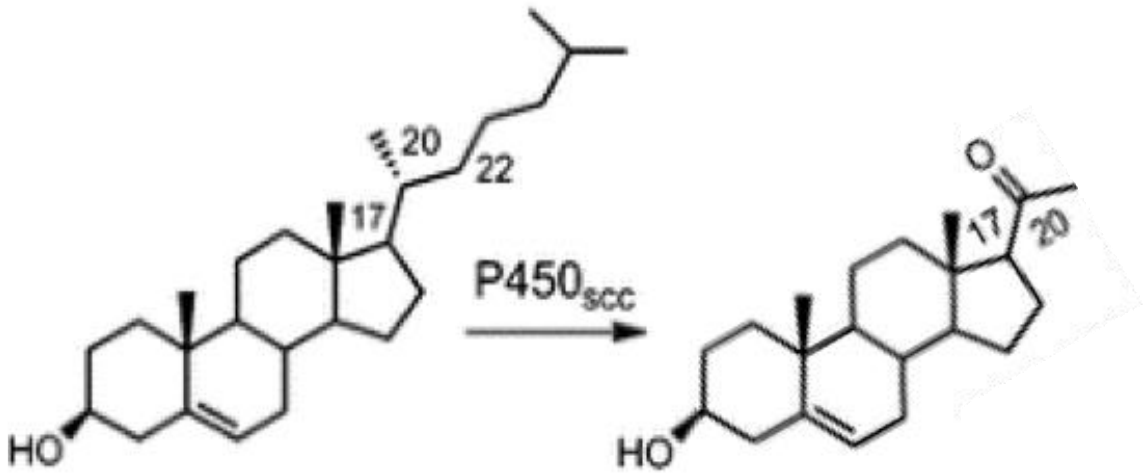


Cholesterol transport to mitochondria



- Cholesterol is anchored in lipid bilayers of plasma and organelle membranes.
- Cholesterol transport to inner mitochondrial membrane mediated by steroid acute regulatory protein (StAR)
- The activity of StAR is generally the rate-limiting step of steroid synthesis pathways and can be modified by signal transduction cascades

Cholesterol side chain cleavage initiates steroidogenesis

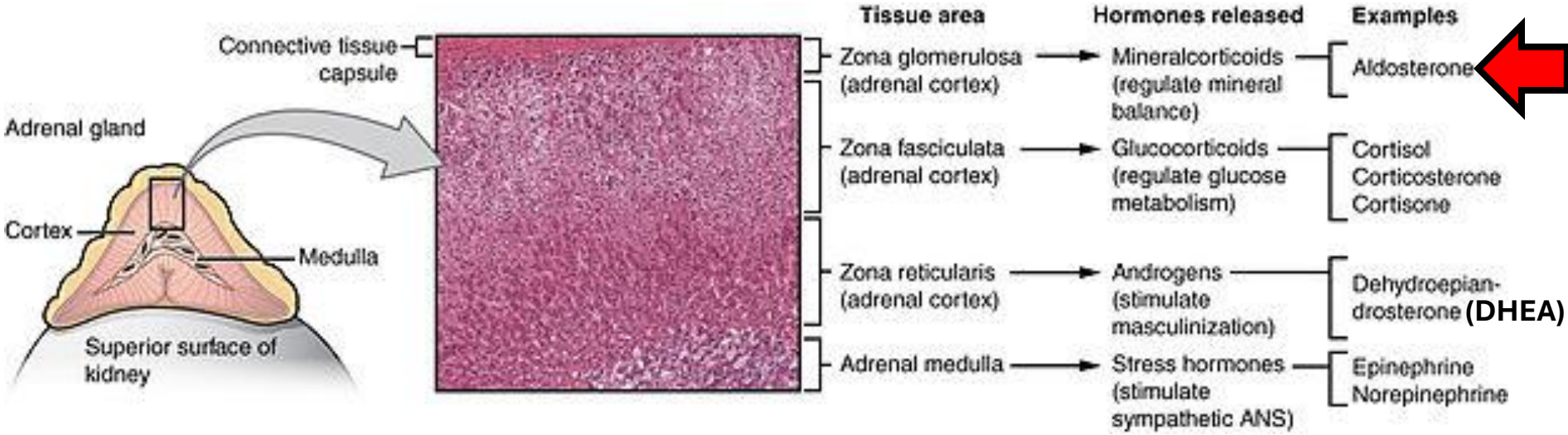


cholesterol
membrane anchored

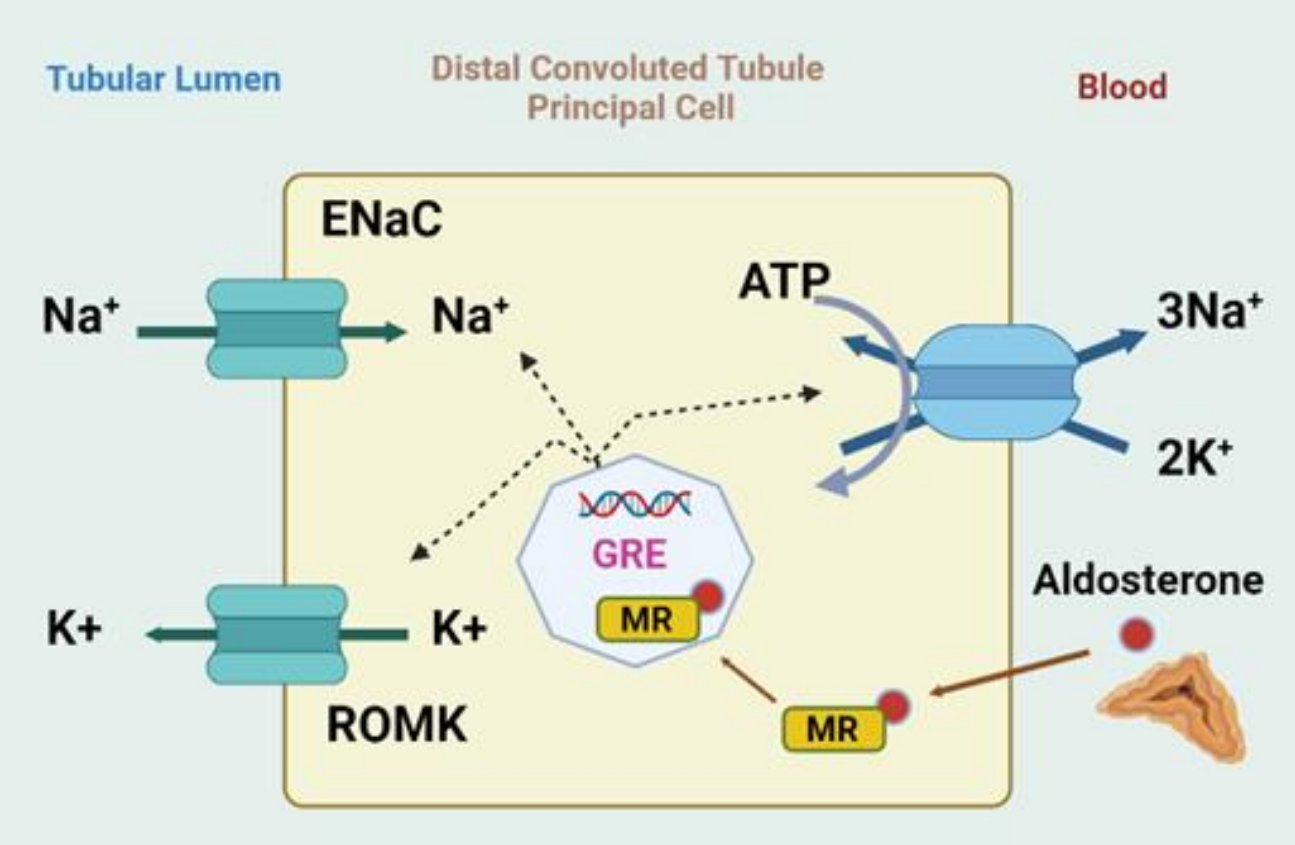
pregnenolone
membrane permeable

- This step carried out by **Cholesterol Side-Chain Cleavage (SCC)** enzyme is common to formation of all steroid hormones.
- Membrane permeability is critical to the function of steroid hormones since nuclear hormone receptors are **intracellular**
- The membrane permeability of steroid hormones means they CANNOT be stored. Regulation of steroid hormone release must be at the level of *de novo synthesis*

Tissue-Specific Steroid Production: Adrenal Glands

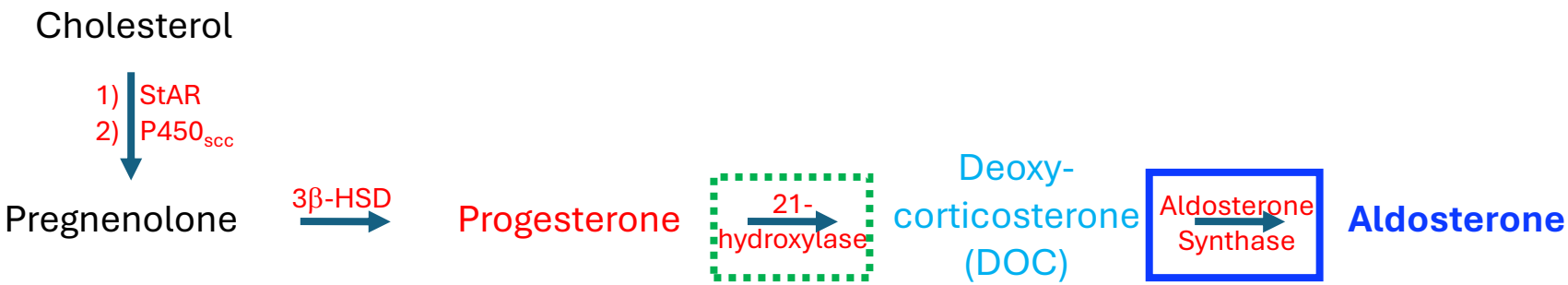


REVIEW: Aldosterone regulates electrolyte balance & blood pressure

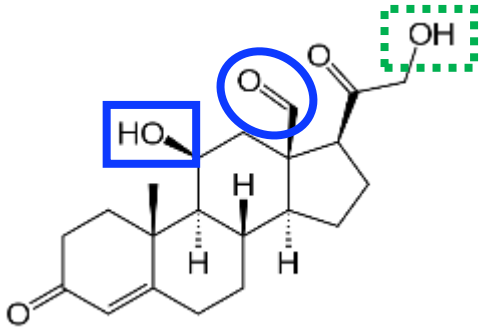


https://ajkdblog.org/2023/02/10/how-does-aldosterone-activation-contribute-to-ckd/

Synthesis of Aldosterone in the Adrenal Zona Glomerulosa

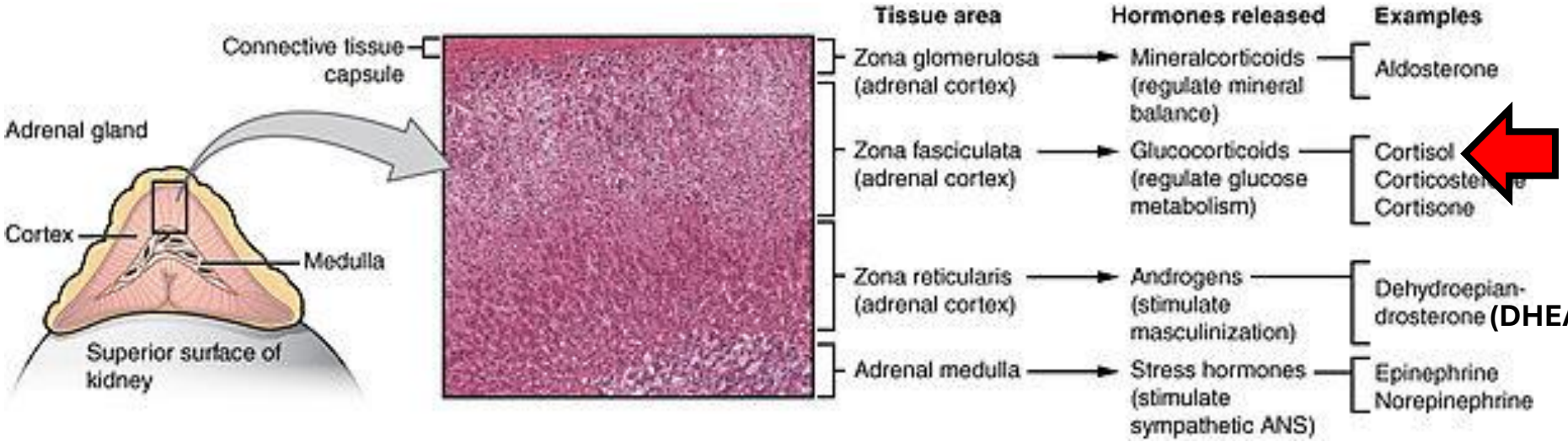


<u>Aldosterone synthase</u> is expressed exclusively in the zona glomerulosa, and is uniquely able to oxidize C18 to an aldehyde (circled in blue)

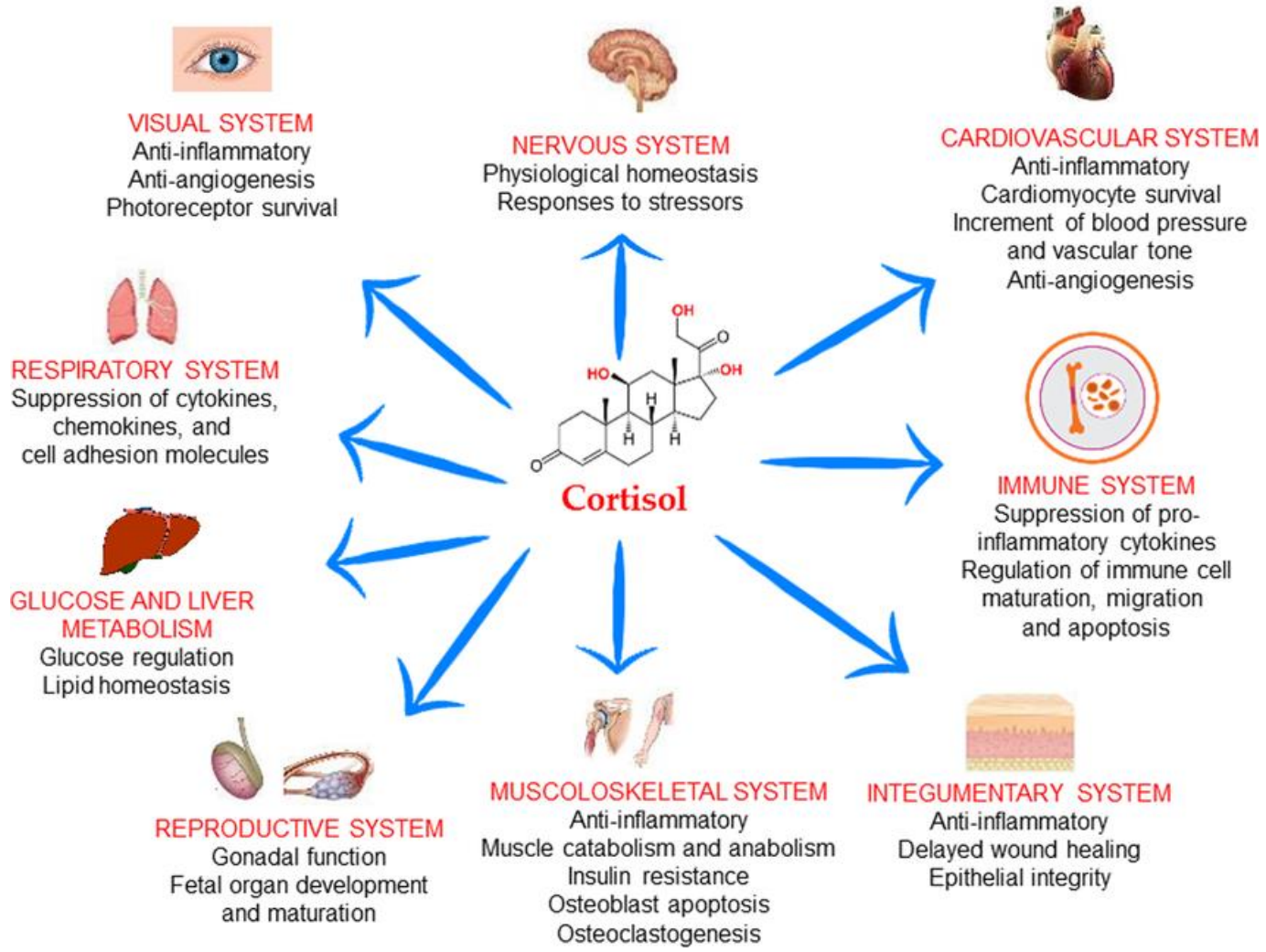


***NOTE**: **progesterone** is an MR antagonist; **DOC** has mineralocorticoid activity*

Tissue-Specific Steroid Production: Adrenal Glands

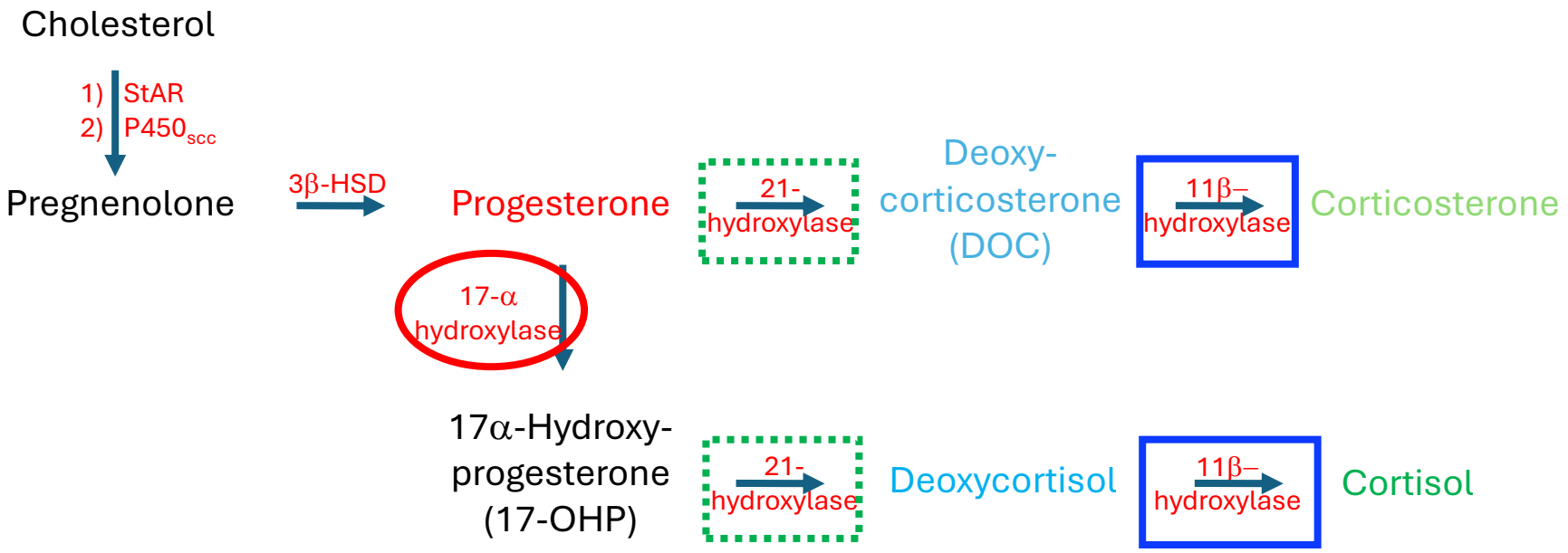


REVIEW: Cortisol regulates metabolism and stress response impacting many organ systems



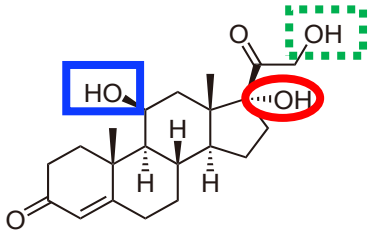
https://www.researchgate.net/figure/Role-of-cortisol-in-health-This-schematic-represents-the-roles-of-glucocorticoids_fig1_347540420

Synthesis of Cortisol in the Adrenal Zona Fasciculata: Intermediates

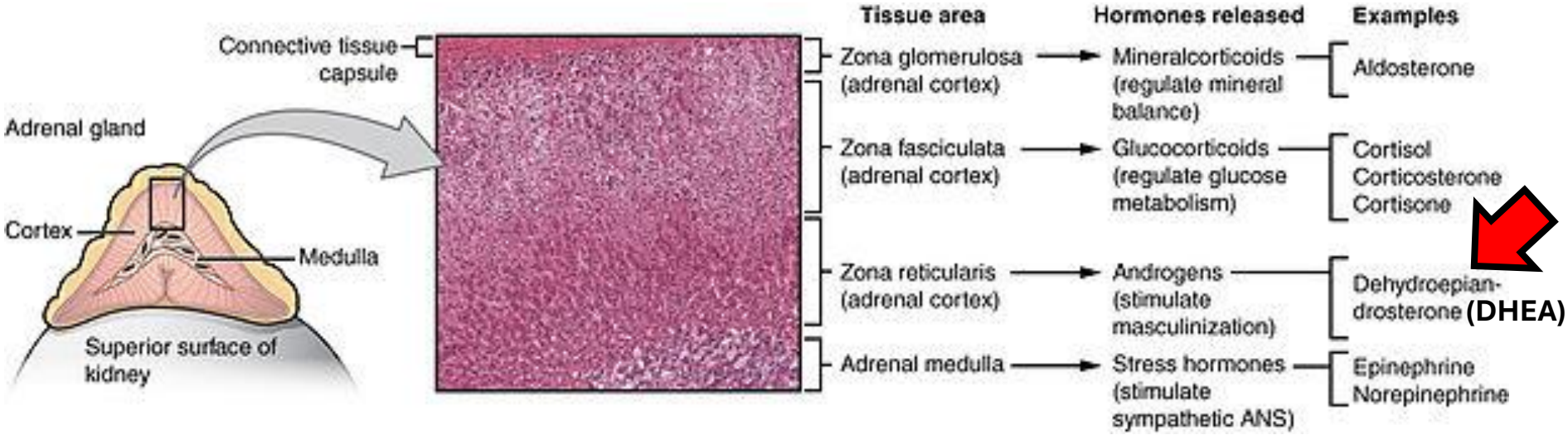


Intermediates with Mineralocorticoid Activity
- -Deoxycorticosterone (minor)
- -Deoxycortisol (minor)

Intermediates with Glucocorticoid Activity
- -Corticosterone (minor)



Tissue-Specific Steroid Production: Adrenal Glands



REVIEW: Dehydroepiandrosterone (DHEA) is an androgen that induces anabolic changes in many systems

Central nervous system

↑ Mood

↓ Anxiety

↓Depression

↑ Cognitive function

↑ Sexual function

Neuro protective

Skin

↑ Sebum production

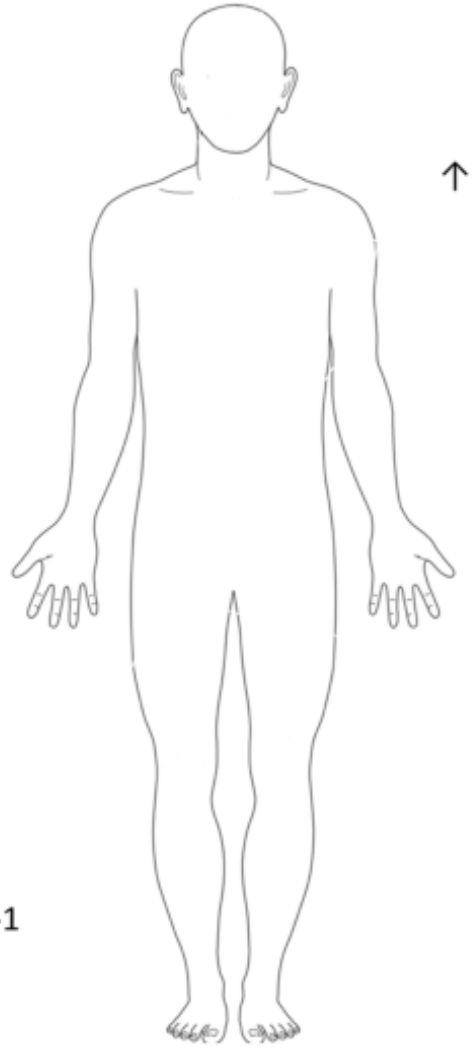
↑ Acne

Metabolism

↑Insulin sensitivity

↓ HDL cholesterol

↑ Insulin-like growth factor-1



Immune system

↓ Plasma cortisol

↑ Neutrophil reactive oxygen species generation

Bone

↑ Bone mineral density

Muscle

↑ Strength

↑ Lean body mass

↓ Fat mass

Other

Anxiolytic

↓ Menopausal thermoregulation disturbances

https://burnstrauma.biomedcentral.com/articles/10.1186/s41038-019-0158-z

Androgen Synthesis in the Adrenal Zona Reticularis

Cholesterol

1) StAR
2) $P450_{scc}$

Pregnenolone —3β-HSD→ Progesterone

17-α hydroxylase ↓ (Pregnenolone)

17-α hydroxylase ↓ (Progesterone)

17α-Hydroxy-pregnenolone —3β-HSD→ 17α-Hydroxy-progesterone (17-OHP)

17-α Hydroxylase + Cytochrome B5 ↓ (17α-Hydroxy-pregnenolone)

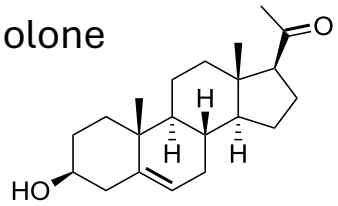
17-α Hydroxylase + Cytochrome B5 ↓ (17α-Hydroxy-progesterone)

Dihydroepi-androsterone (DHEA) —3β-HSD→ Androstenedione

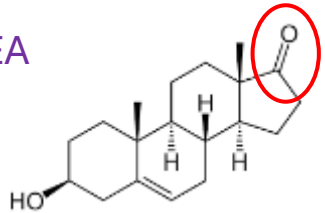
21 Carbon

19 Carbon

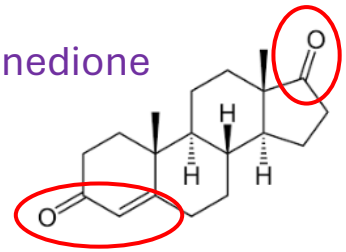
Pregnenolone



DHEA

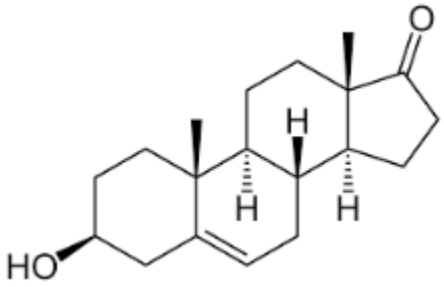


Androstenedione

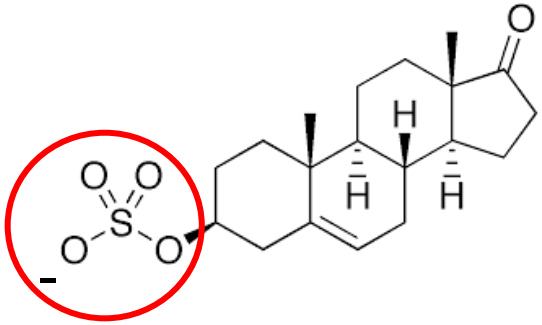


Androgen Synthesis in the Adrenal Zona Reticularis: DHEA and DHEAS

- DHEA and androstenedione
  - Weak androgen activity (low affinity for androgen receptor)
  - Can be converted to strong androgens (testosterone and DHT) in peripheral tissues (e.g., skin, hair follicles)
  - Weak estrogen activity, but readily transformed into potent estrogens in peripheral tissues (e.g., vagina)



Sulfotransferase



- DHEAS
  - Found at 100-fold higher levels than DHEA
    - Higher solubility than DHEA
    - Tighter binding to albumin → 10-20-fold longer serum half-life than DHEA
  - Does not activate steroid hormone receptors
  - Converted to DHEA in peripheral tissues through activity of sulfatases

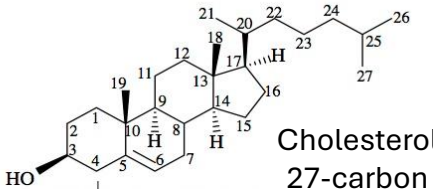
Androgen Synthesis in the Adrenal Zona Reticularis

- DHEAS is the most abundant circulating corticosteroid
  - 5-10-fold higher than cortisol; 100-500 fold higher than testosterone (before puberty); 1000-10,000-fold higher than estradiol (pre/post menopause)
  - In healthy adults, adrenals produce 15 mg total of DHEAS per day.
- Levels vary throughout life
  - High in neonate (from fetal zone of fetal adrenal gland)
  - Low in childhood
  - Start to rise at age 7-8 (adrenarche)
  - Peak at age 20-30
  - Slowly decline with age
- Source of most estrogens in post-menopausal women

Tissue-Specific Steroidogenic Pathways

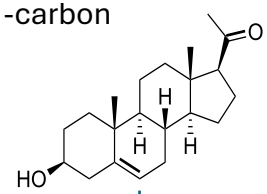
	Enzymes Expressed	Important Intermediates	Major Steroid Produced
Zona glomerulosa	Cholesterol SCC 3βHSD 21-Hydroxylase **Aldosterone Synthase**	Progesterone DOC	**Aldosterone**
Zona fasciculata	Cholesterol SCC 3βHSD 21-Hydroxylase **11β-hydroxylase** **17α-Hydroxylase**	Progesterone **17-OHP** DOC Corticosterone	**Cortisol**
Zona reticularis	Cholesterol SCC 3βHSD **17α-Hydroxylase** **Cytochrome b5** **Sulfotransferase**	Progesterone **17-OHP**	**DHEA** **DHEAS** **Androstenedione**

Steroidogenesis Overview

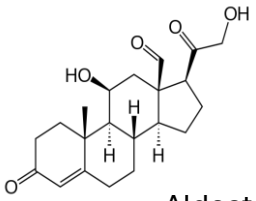


Cholesterol
27-carbon

Pregnenolone
21-carbon



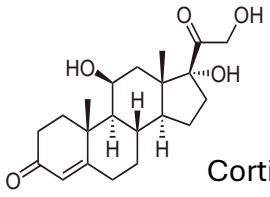
3βHSD 21-Hydroxylase **Aldosterone Synthase**



Aldosterone
21-carbon

17α-Hydroxylase

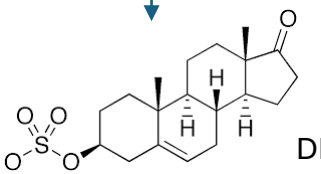
3βHSD 21-Hydroxylase **11β-hydroxylase**



Cortisol
21-carbon

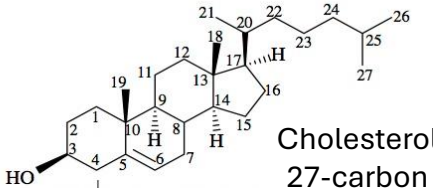
17α-Hydroxylase +
Cytochrome b5

Sulfotransferase



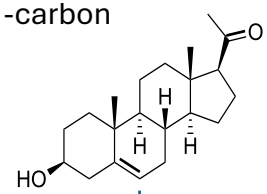
DHEAS
19-carbon

Clinical Correlation: Steroidogenic Enzyme Deficiency



Cholesterol
27-carbon

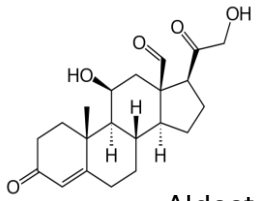
Pregnenolone
21-carbon



3βHSD

21-Hydroxylase

Aldosterone Synthase



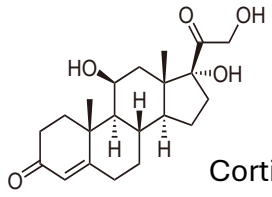
Aldosterone
21-carbon

17α-Hydroxylase

3βHSD

21-Hydroxylase

11β-hydroxylase

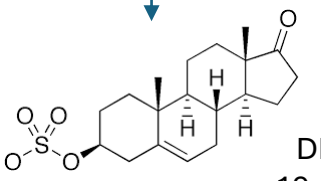


Cortisol
21-carbon

17α-Hydroxylase +
Cytochrome b5

Sulfotransferase

21-Hydroxylase Deficiency



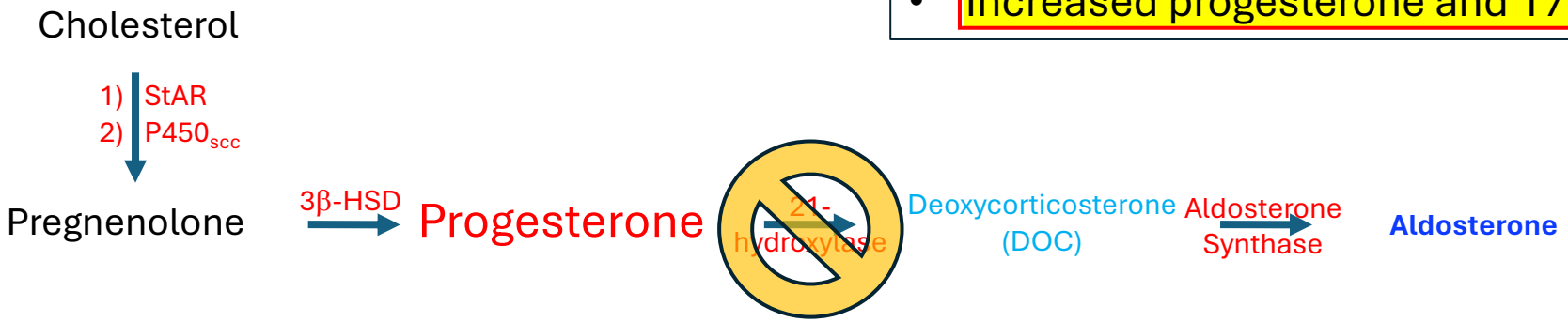
DHEAS
19-carbon

Clinical Correlation: Steroidogenic Enzyme Deficiency

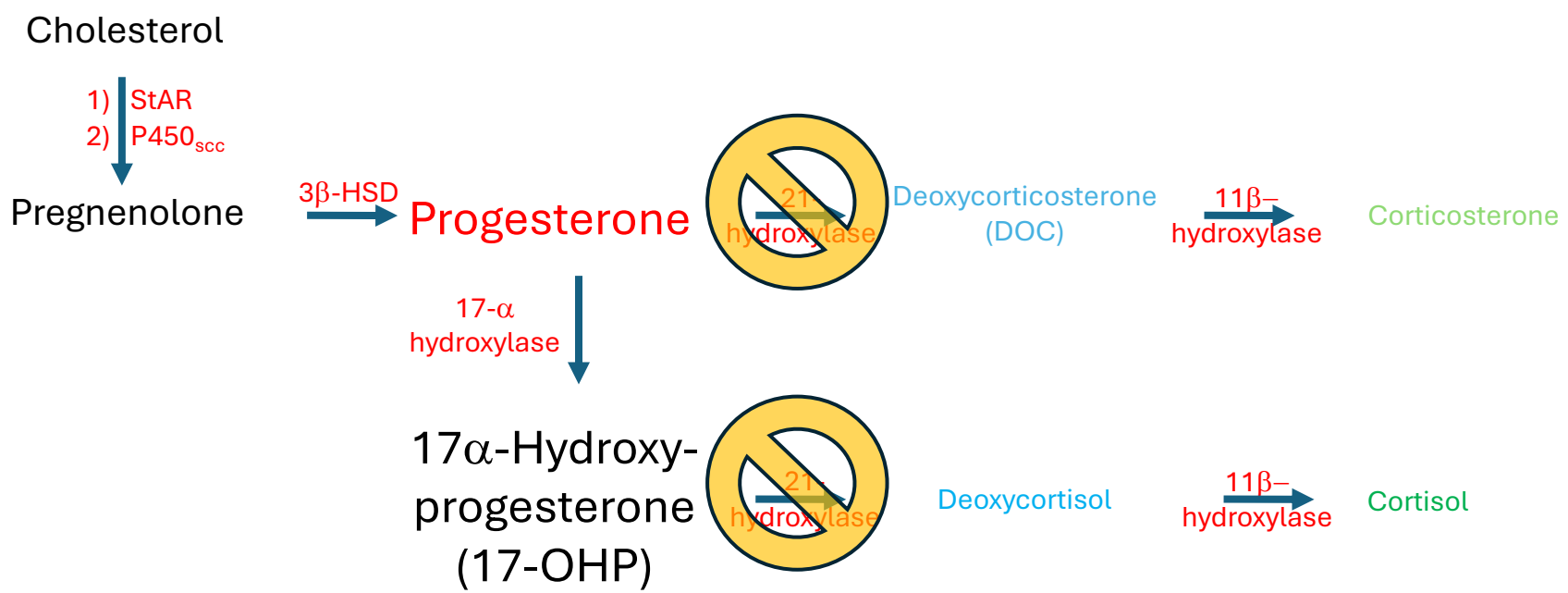
21-Hydroxylase Deficiency

- Decreased aldosterone or cortisol
- Increased progesterone and 17-OHP

Zona Glomerulosa



Zona Fasciculata

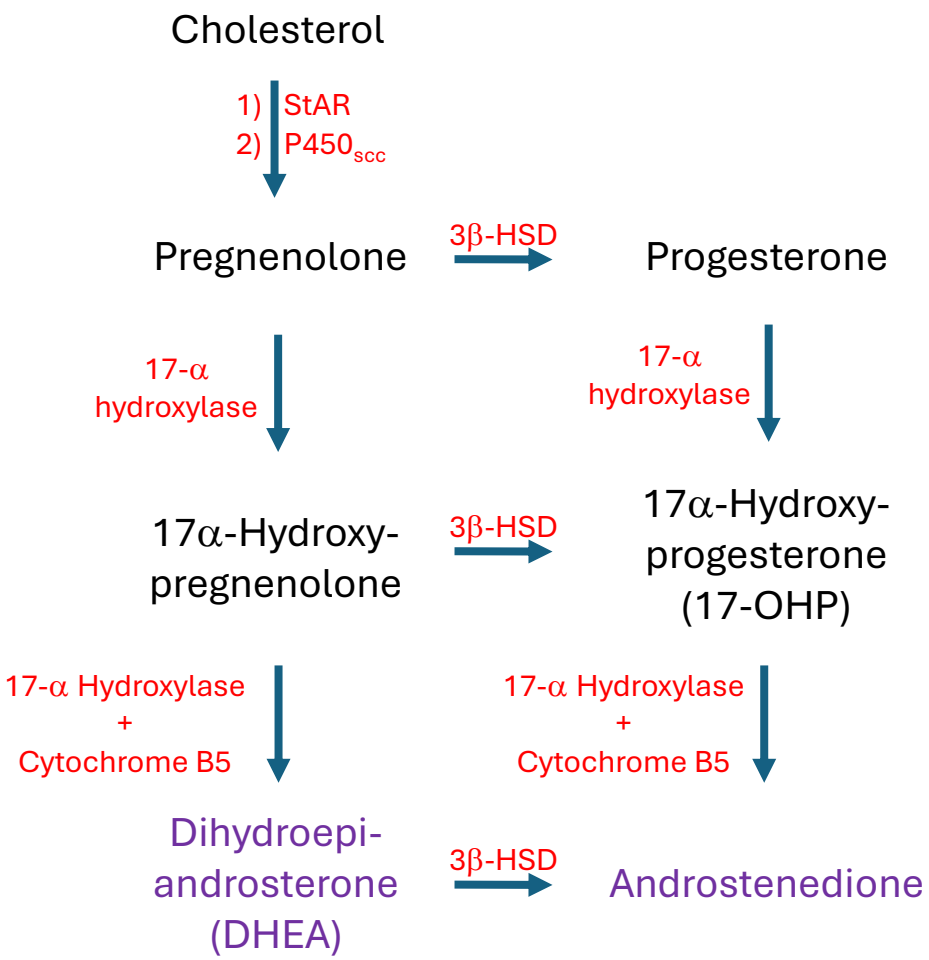


Clinical Correlation: Steroidogenic Enzyme Deficiency

21-Hydroxylase Deficiency

- Decreased aldosterone or cortisol
- Increased progesterone and 17-OHP

Zona Reticularis



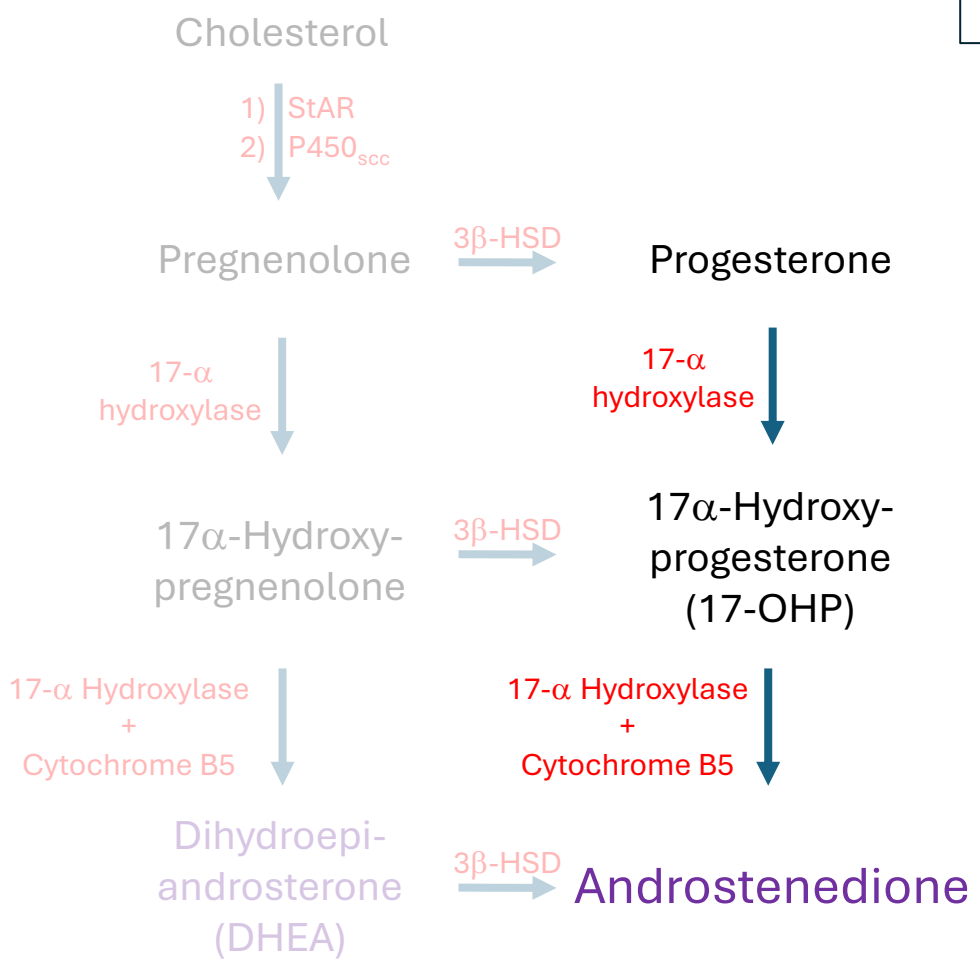
21-Hydroxylase is not expressed in the Zona reticularis, but....

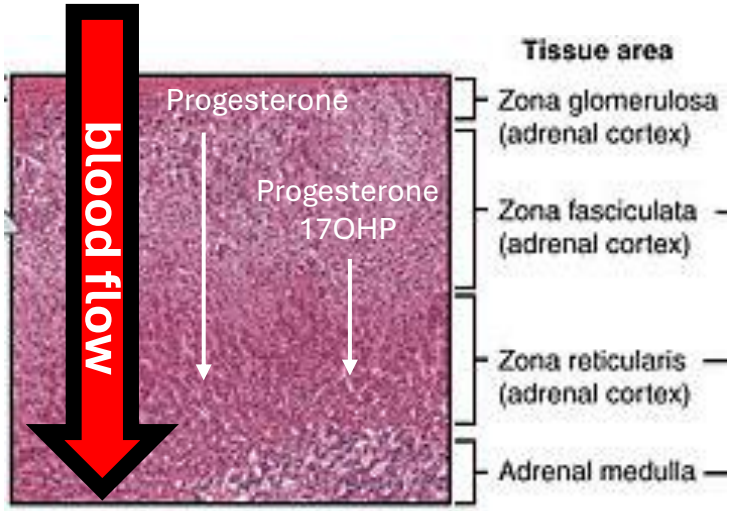
Clinical Correlation: Steroidogenic Enzyme Deficiency

21-Hydroxylase Deficiency

- Decreased aldosterone or cortisol
- Increased progesterone and 17-OHP
- Increased androstenedione

Zona Reticularis





Clinical Correlation: Steroidogenic Enzyme Deficiency

21-Hydroxylase Deficiency

- Decreased aldosterone
  - Electrolyte imbalance: hyponatremia, hyperkalemia
  - Dehydration
  - Hypotension
- Decreased cortisol
  - Reduced stress response and impaired metabolism
  - Lethargy
- Increased progesterone (MR antagonist)
  - Contributes to electrolyte imbalance
- Increased androstenedione
  - Impaired secondary sexual development (e.g., virilization of XX)

But…the impact of these effects are greatly enhanced by an increase in adrenal mass due to disrupted feedback regulation of adrenal hormones!

Learning Objectives

1. Map the pathways involved in synthesis of adrenal steroids and explain how tissue-specific production of corticosteroids is achieved.

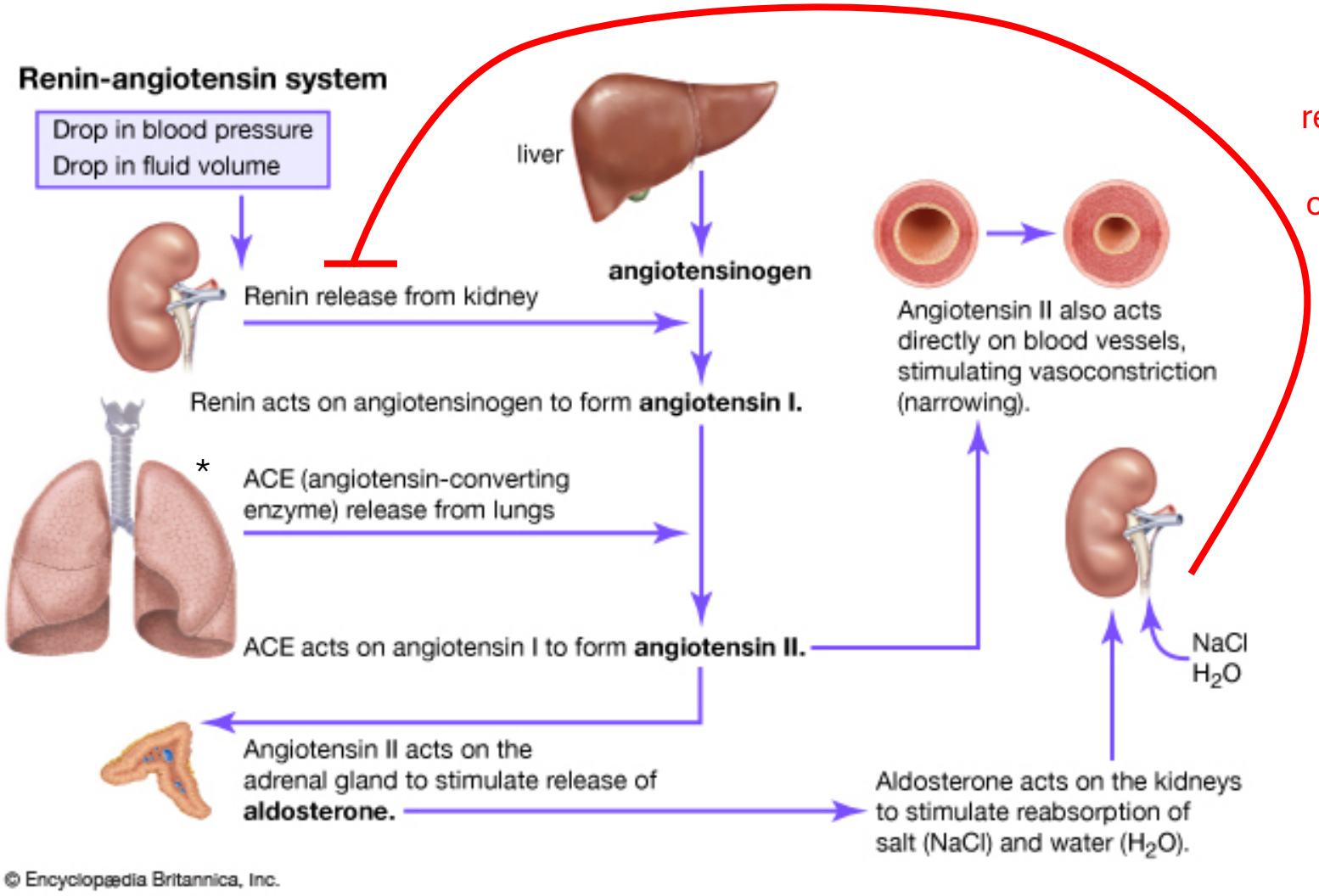
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5. Outline common laboratory tests for identifying patients with defects in steroidogenesis and explain how they discriminate between 21-hydroxylase deficiency, 11β-hydroxylase deficiency and 17α-hydroxylase deficiency.

Regulation of Aldosterone Production
Renin-Angiotensin-Aldosterone System

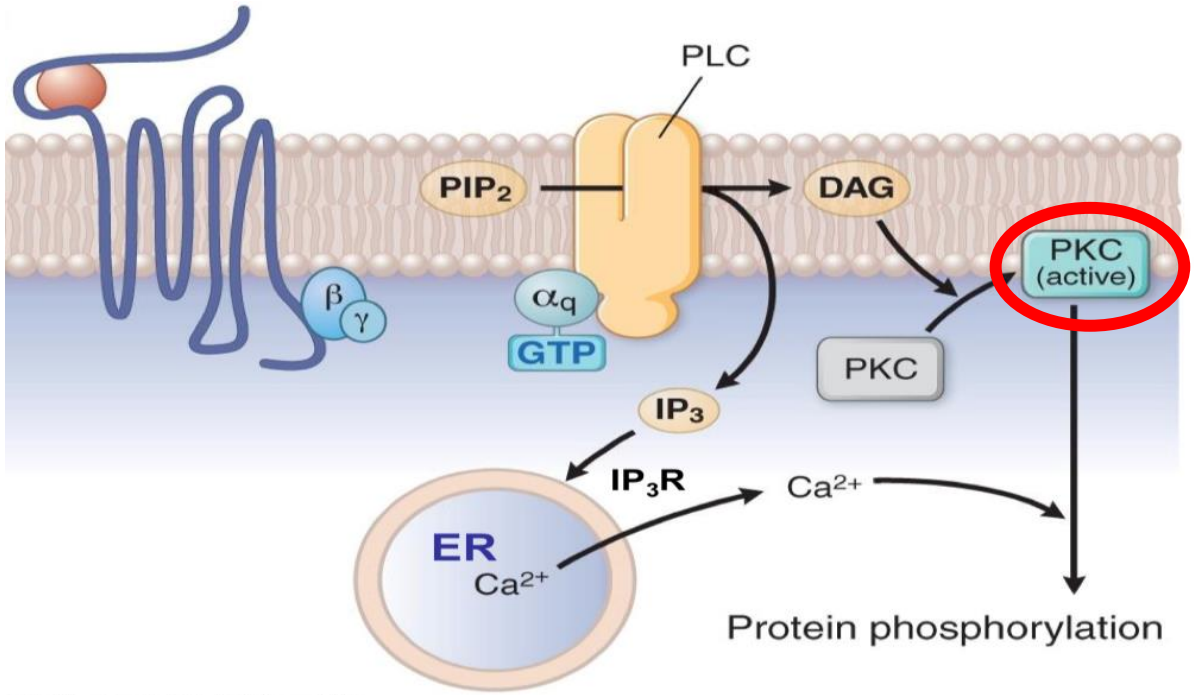


Aldosterone "feedback" on renin production is indirect (via change in blood pressure)

*ACE2 (the receptor for SARS-CoV2) converts angiotensin II (vasoconstrictor) to angiotensin (vasodilator)

Regulation of Aldosterone Production: Actions of Angiotensin II

- Angiotensin II receptor (AGTR1)
  - **Expressed in zona glomerulosa, but <u>NOT</u> in zona fasciculata**
  - G-protein coupled receptor coupled to Gαq/11 protein
  - Activated (GTP-bound) Gq/11 activates phospholipase C, which increases cytosolic calcium and activates protein kinase C (PKC)



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Regulation of Aldosterone Production: Actions of Angiotensin II

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  - G-protein coupled receptor coupled to Gαq/11 protein
  - Activated (GTP-bound) Gq/11 activates phospholipase C, which increases cytosolic calcium and activates protein kinase C (PKC)

- AGTR1 stimulation
  - Short-term (minutes) – increases rate of aldosterone synthesis via increased StAR activity (through phosphorylation)
  - Intermediate-term (hours to days) – increases cellular levels of enzymes, especially **Aldosterone Synthetase**, and other proteins involved in aldosterone synthesis
  - Long-term (weeks) – increases cell mass of Zona glomerulosa

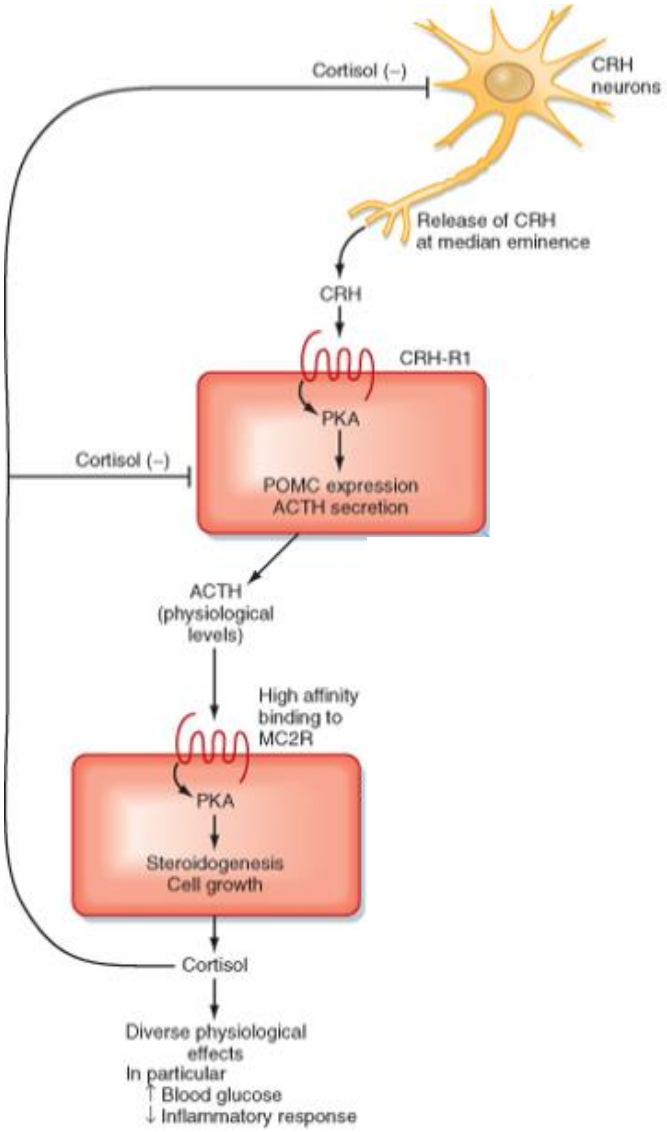
Clinical Correlation: Steroidogenic Enzyme Deficiency

21-Hydroxylase Deficiency

- Decreased aldosterone
  - Electrolyte imbalance: hyponatremia, hyperkalemia
  - Dehydration
  - Hypotension
- Decreased cortisol
  - Reduced stress response and impaired metabolism
  - Lethargy
- Increased progesterone (MR antagonist)
  - Contributes to electrolyte imbalance
- Increased androstenedione
  - Impaired secondary sexual development (e.g., virilization of females)
- Sustained increased renin levels → increased Zona glomerulosa mass (more progesterone produced)

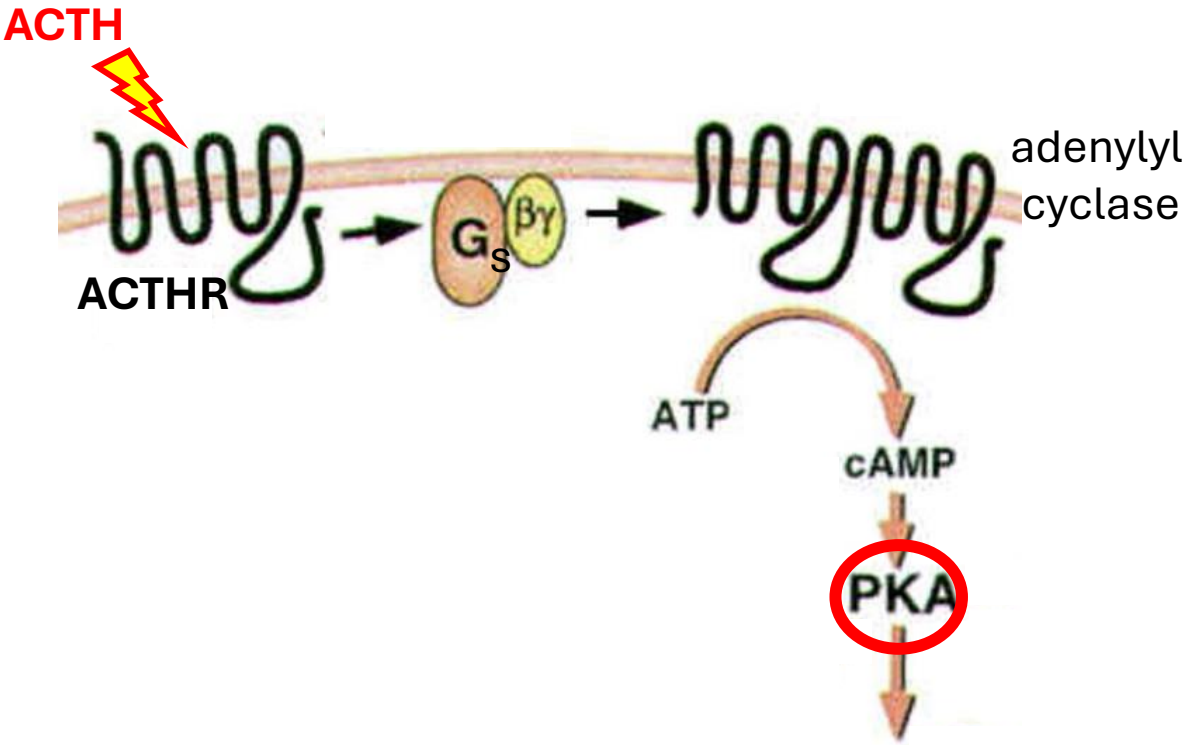
Regulation of Cortisol Production: Hypothalamic-Pituitary-Adrenal Axis

Cortisol feeds back **directly** to inhibit CRH & ACTH secretion



Regulation of Cortisol Production: Actions of ACTH

- ACTH receptor (ACTHR)
  - **Expressed in zona fasciculata, but <u>NOT</u> in zona glomerulosa**
  - G-protein coupled receptor coupled to Gαs
  - Signaling cascade through cAMP and protein kinase A (PKA)



Regulation of Cortisol Production: Actions of ACTH

- ACTH receptor (ACTHR)
  - **Expressed in zona fasciculata, but <u>NOT</u> in zona glomerulosa**
  - G-protein coupled receptor coupled to Gαs
  - Signaling cascade through cAMP and protein kinase A (PKA)

- ACTHR stimulation
  - Short-term (minutes) – increases rate of cortisol synthesis via increased StAR activity (through phosphorylation)
  - Intermediate-term (hours to days) – increases cellular levels of enzymes and other proteins involved in cortisol synthesis
  - Long-term (weeks) – increases cell mass of Zona fasciculata

Clinical Correlation: Steroidogenic Enzyme Deficiency

21-Hydroxylase Deficiency

- Decreased aldosterone
  - Electrolyte imbalance: hyponatremia, hyperkalemia
  - Dehydration
  - Hypotension
- Decreased cortisol
  - Reduced stress response and impaired metabolism
  - Lethargy
- Increased progesterone (MR antagonist)
  - Contributes to electrolyte imbalance
- Increased androstenedione
  - Impaired secondary sexual development (e.g., virilization of females)
- Sustained increased renin levels → increased Zona glomerulosa mass (more progesterone produced)
- Sustained increased ACTH levels → increased Zona fasciculata mass (more progesterone and 17OHP produced)

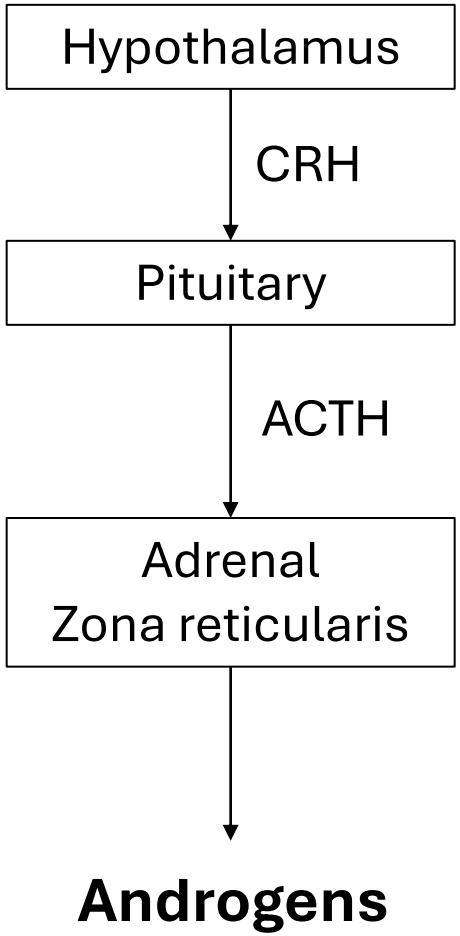
Regulation of Adrenal Androgen Production: Actions of ACTH

- Overall, the regulation of steroidogenesis in the zona reticularis is incompletely understood, but these cells contain the ACTH receptor.

- ACTH receptor (ACTHR)
  - **Expressed in zona reticularis**
  - G-protein coupled receptor coupled to Gαs
  - Signaling cascade through cAMP and protein kinase A (PKA)

- ACTHR stimulation
  - Short-term (minutes) – increases rate of DHEA and androstenedione synthesis via increased StAR activity (through phosphorylation)
  - Intermediate-term (hours to days) – increases cellular levels of enzymes and other proteins involved in adrenal androgen synthesis
  - Long-term (weeks) – increases cell mass of Zona reticularis

Regulation of Adrenal Androgen Production



Note: No feedback regulation on CRH/ACTH!

Clinical Correlation: Steroidogenic Enzyme Deficiency

21-Hydroxylase Deficiency

- Decreased aldosterone
  - Electrolyte imbalance: hyponatremia, hyperkalemia
  - Dehydration
  - Hypotension
- Decreased cortisol
  - Reduced stress response and impaired metabolism
  - Lethargy
- Increased progesterone (MR antagonist)
  - Contributes to electrolyte imbalance
- Increased androstenedione
  - Impaired secondary sexual development (e.g., virilization of females)
- Sustained increased renin levels → increased Zona glomerulosa mass (more progesterone produced)
- Sustained increased ACTH levels → increased Zona fasciculata mass (more progesterone and 17OHP produced)
- Sustained increased ACTH levels → increased Zona reticularis mass (more DHEA and androstenedione produced)

Clinical Correlation: Steroidogenic Enzyme Deficiency

21-Hydroxylase Deficiency

- Decreased aldosterone
  - Electrolyte imbalance: hyponatremia, hyperkalemia
  - Dehydration
  - Hypotension
- Decreased cortisol
  - Reduced stress response and impaired metabolism
  - Lethargy
- Increased progesterone (MR antagonist)
  - Contributes to electrolyte imbalance
- Increased androstenedione
  - Impaired secondary sexual development (e.g., virilization of females)
- Sustained increased renin levels → increased Zona glomerulosa mass (more progesterone produced)
- Sustained increased ACTH levels → increased Zona fasciculata mass (more progesterone and 17OHP produced)
- Sustained increased ACTH levels → increased Zona reticularis mass (more DHEA and androstenedione produced)

Adrenal hyperplasia
If from primary adrenal steroidogenic enzyme deficiency, **congenital adrenal hyperplasia (CAH)**

Summary

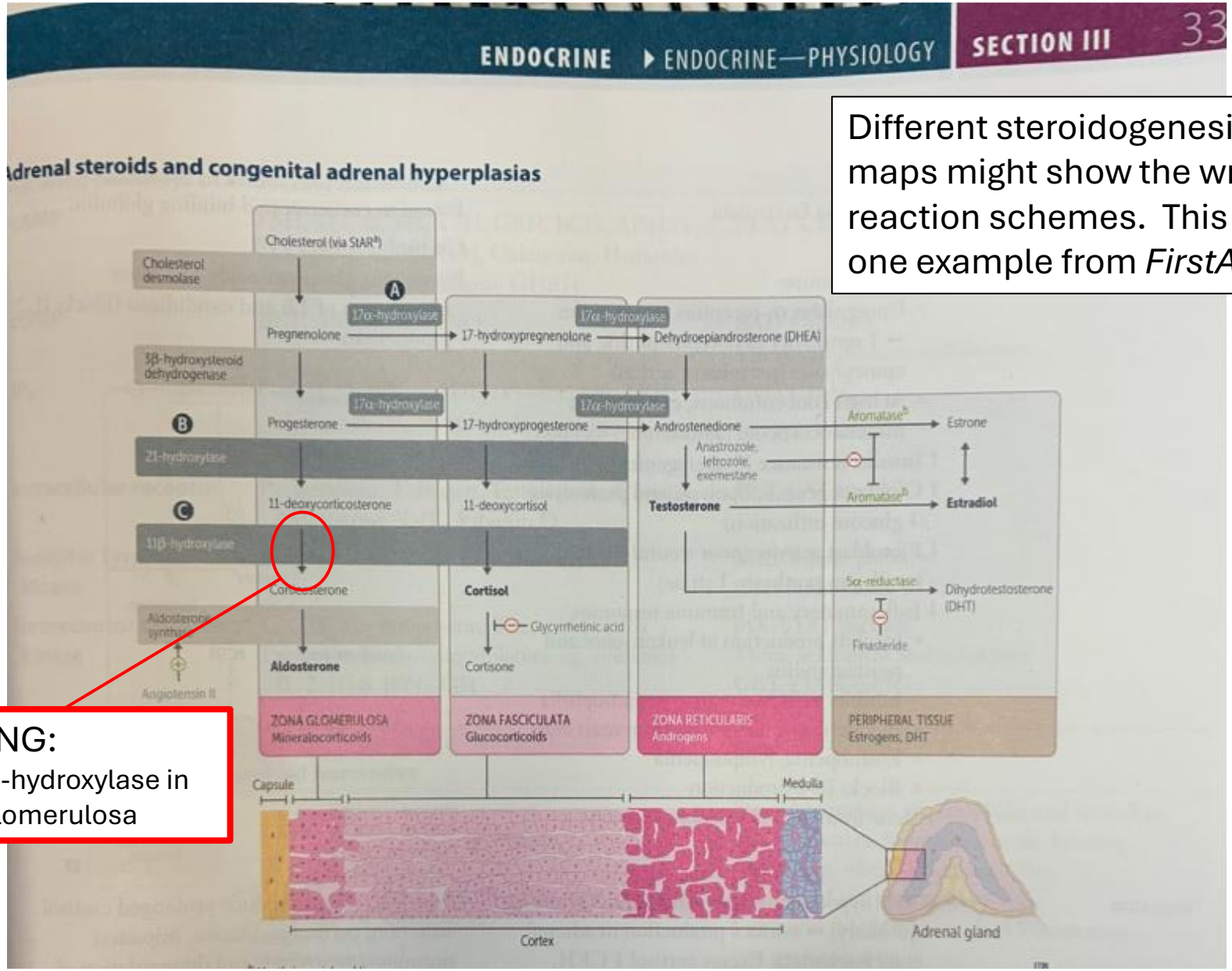
1. Tissue-specific corticosteroid production is enforced by restricted expression of GPCRs (AGTR1/ACTHR) and steroidogenic enzymes (e.g., aldosterone synthase, 17α-hydroxylase, 11β-hydroxylase) and cofactors (cytochrome b5).
2. Aldosterone synthesis is regulated by RAAS via AGTII. Cortisol synthesis is regulated by the HPA axis via ACTH. DHEA synthesis is regulated by ACTH without any feedback inhibition.
3. An enzyme deficiency along a steroidogenic pathway in the adrenals leads to both a loss of final steroid product and a buildup of steroid precursors that can either have innate activity or be processed into steroids with activity.

    Example – 21-Hydroxylase Deficiency → decreased cortisol, aldosterone; increased progesterone, 170HP; increased adrenal androgen production.

4. Chronic stimulation of GPCR receptors (AGTR1, ACTHR) leads to congenital adrenal hyperplasia

BREAK

Steroidogenesis Overview



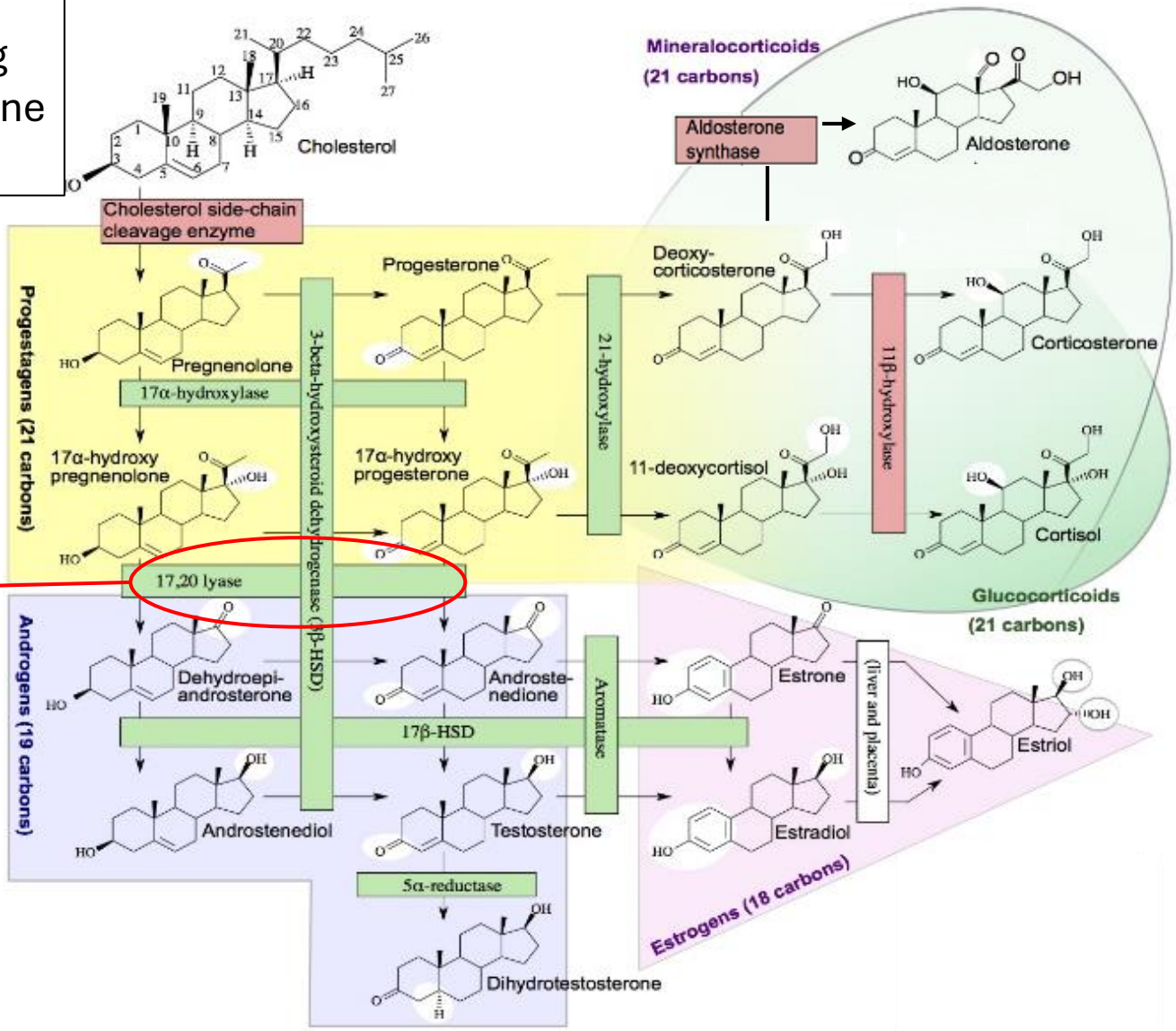
Different steroidogenesis maps might show the wrong reaction schemes. This is one example from *FirstAid*.

WRONG:
No 11β-hydroxylase in Zona glomerulosa

Steroidogenesis Overview

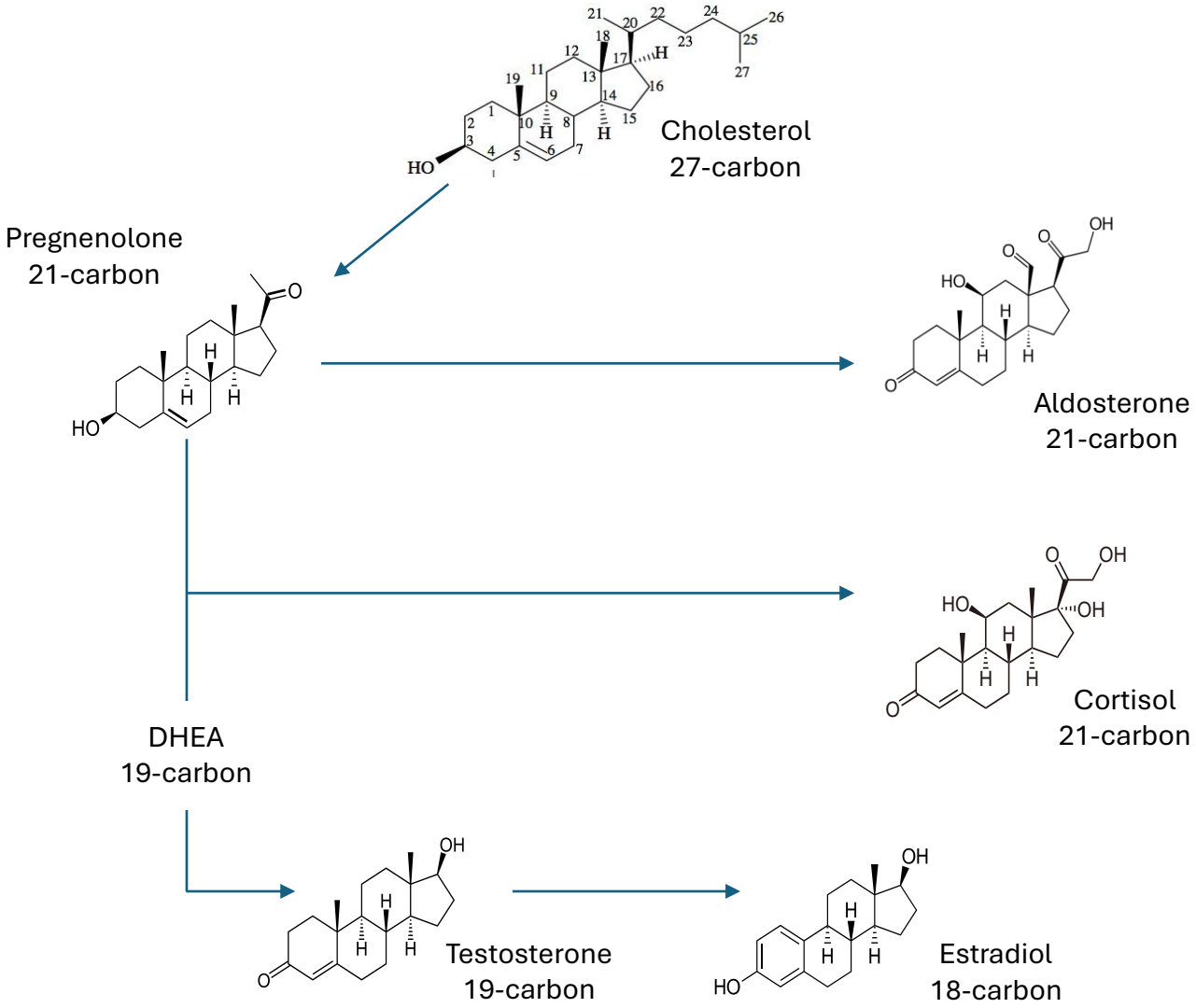
Different steroidogenesis maps might show the wrong reaction schemes. This is one example from the internet.

WRONG:
There is no enzyme called 17,20 lyase – the reaction is actually catalyzed by 17α-hydroxylase when cofactor b5 is present.



Steroidogenesis Overview

- In no cell do all of these reactions happen.
- Expression patterns of enzymes and cofactors/enzyme modifiers determine tissue-specific steroid production.
- Intermediates can diffuse between cells to complete synthesis



Learning Objectives

1. Map the pathways involved in synthesis of adrenal steroids and explain how tissue-specific production of corticosteroids is achieved.

2. Describe the hormonal regulation of adrenal steroid production and explain why chronic signaling leads to congenital adrenal hyperplasia (CAH).

3. Map the pathways involved in the synthesis of testosterone, estrogen and progesterone in gonadal tissue and describe their hormonal regulation.

4. Compare and contrast the effects on blood pressure, metabolism, stress response, and sexual development of common disorders of steroidogenesis (21-hydroxylase deficiency, 11β-hydroxylase deficiency, 17α-hydroxylase deficiency, 11β-hydroxysteroid dehydrogenase deficiency) and describe their pathophysiological mechanisms.

5. Outline common laboratory tests for identifying patients with defects in steroidogenesis and explain how they discriminate between 21-hydroxylase deficiency, 11β-hydroxylase deficiency and 17α-hydroxylase deficiency.

Androgen Synthesis in the Adrenal Zona Reticularis

Cholesterol

1) StAR
2) $P450_{scc}$

Pregnenolone —3β-HSD→ Progesterone

17-α hydroxylase ↓ (Pregnenolone)

17-α hydroxylase ↓ (Progesterone)

17α-Hydroxy-pregnenolone —3β-HSD→ 17α-Hydroxy-progesterone (17-OHP)

17-α Hydroxylase + Cytochrome B5 ↓ (17α-Hydroxy-pregnenolone)

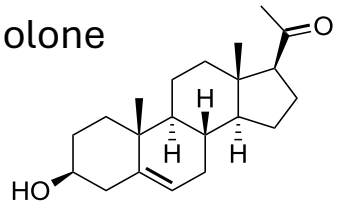
17-α Hydroxylase + Cytochrome B5 ↓ (17α-Hydroxy-progesterone)

Dihydroepi-androsterone (DHEA) —3β-HSD→ Androstenedione

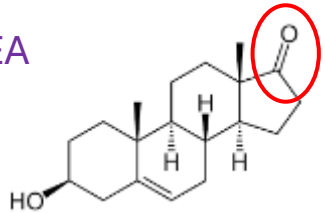
21 Carbon

19 Carbon

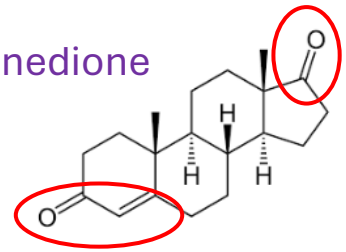
Pregnenolone



DHEA



Androstenedione



Testosterone Synthesis in the Testes

Cholesterol —(Side chain cleavage)→ Pregnenolone —(3βHSD)→ Progesterone

Pregnenolone —(17α-hydroxylase)→ 17α-Hydroxy-Pregnenolone —(3βHSD)→ 17OHP

Progesterone —(17α-hydroxylase)→ 17OHP

17α-Hydroxy-Pregnenolone —(17α-hydroxylase + cytochrome b5)→ DHEA —(3βHSD)→ Androstenedione

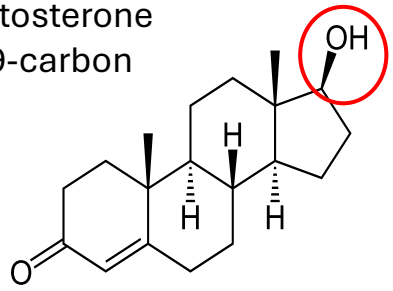
17OHP —(17α-hydroxylase + cytochrome b5)→ Androstenedione

Common steroid synthetic steps in both adrenal zona reticularis and testes

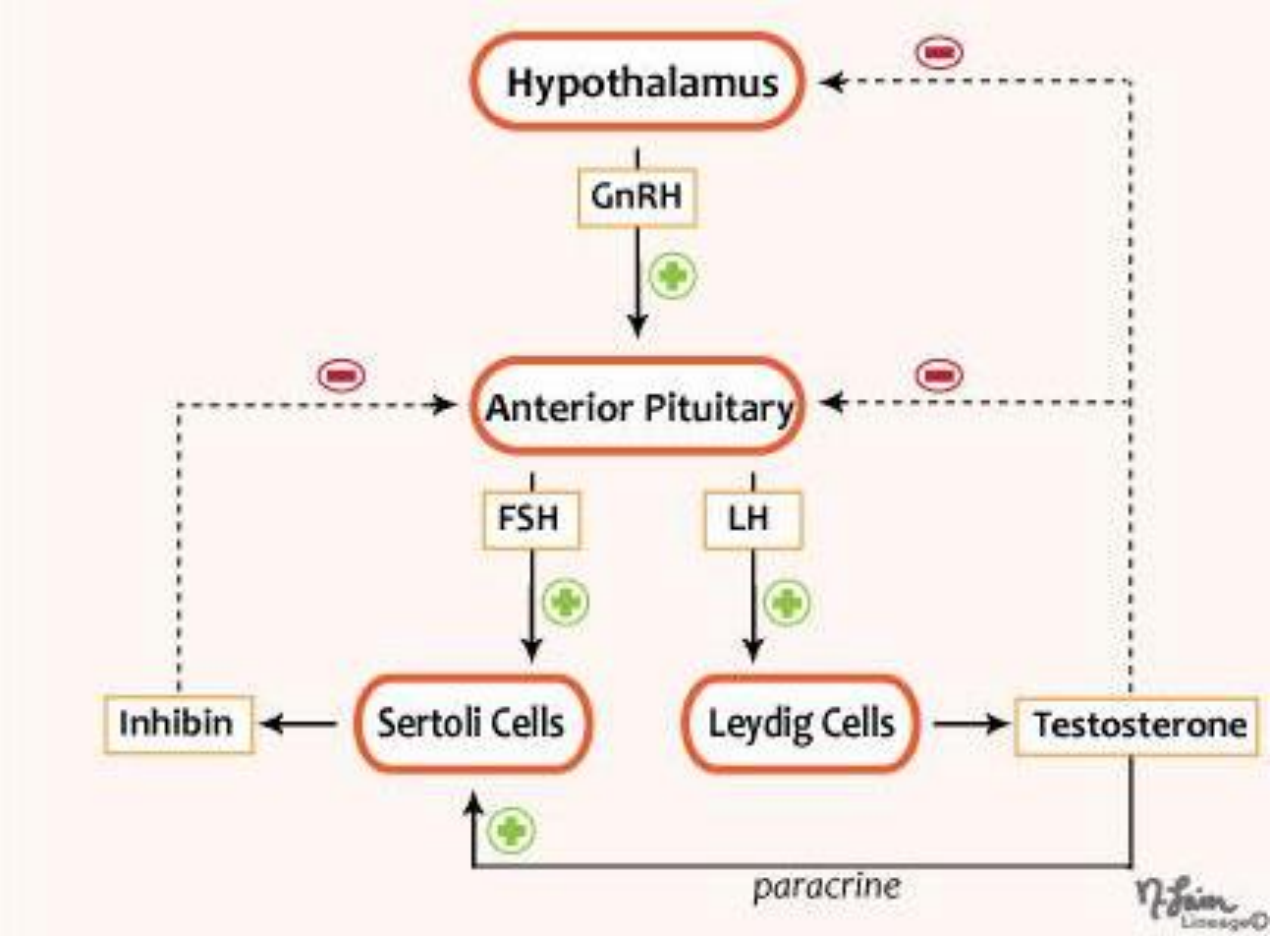
DHEA —(17βHSD)→ Androstenediol —(3βHSD)→ Testosterone

Androstenedione —(17βHSD)→ Testosterone

Testosterone
19-carbon



Regulation of Testosterone Production: Hypothalamic-Pituitary-Testes Axis



GnRH Rec
GPCR (Gq/11)

LH Rec
GPCR (Gs)

FSH Rec
GPCR (Gs)

Testosterone Rec
NHR

Inhibin blocks FSH production

Regulation of Leydig Cell Testosterone Production: Actions of LH

- LH receptor
  - G protein-coupled receptor coupled to Gs protein
  - **Expressed in <u>Leydig cells</u>**
  - Signaling cascade through cAMP and PKA

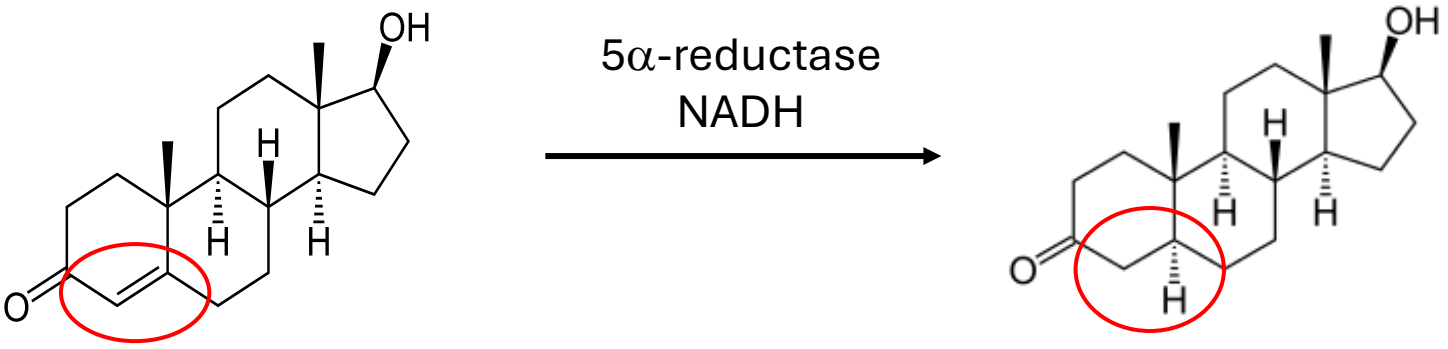
- LH stimulation
  - Short-term (minutes) – increases rate of testosterone synthesis through activity of StAR
  - Intermediate-term (days) – increases cellular levels of enzymes and other proteins involved in cholesterol bioavailability and testosterone synthesis
  - Long-term (weeks) – stimulates some proliferation of Leydig cell progenitors. (Regulation from FSH-stimulated Sertoli cells is pleiotropic).

Regulation of Sertoli Cell Function: Actions of FSH

- Sertoli Cell function
  - Form blood-testis barrier
  - Support and nourish spermatozoa
  - Converts androgens to estrogens via aromatase
- FSH receptor
  - G protein-coupled receptor coupled to Gs protein
  - **Expressed in <u>Sertoli cells</u>**
  - Signaling cascade through cAMP and PKA
- FSH stimulation
  - Secrete inhibin B → inhibits FSH secretion in pituitary
  - Secrete androgen-binding protein (maintain local levels of testosterone in seminiferous tubules)
  - Cell proliferation → increases spermatogenic capacity

Tissue-Specific Steroid Modification: Testosterone Signaling

- **5α-Reductase**
  - 2 isoforms – one in peripheral tissue (e.g., hair follicles) and another predominantly in male reproductive tissue (e.g., prostate)
  - Hydroxysteroid dehydrogenase – catalyzes reduction of the D4 double bond in testosterone to form dihydrotestosterone (DHT).
    - DHT is 5-10-fold more potent at activating the adrenergic receptor than testosterone
  - Also utilized in the degradative pathway for 21-carbon steroids such as cortisol, progesterone, 17OHP, etc.



Testosterone

Dihydrotestosterone (DHT)

Estrogen Synthesis in the Ovaries

Cholesterol

1) StAR
2) $P450_{scc}$

Pregnenolone —3βHSD→ Progesterone

17α-hydroxylase ↓ (Pregnenolone) | 17α-hydroxylase ↓ (Progesterone)

17α-Hydroxy-Pregnenolone —3βHSD→ 17OHP

17α-hydroxylase + cytochrome b5 ↓ (17α-Hydroxy-Pregnenolone) | 17α-hydroxylase + cytochrome b5 ↓ (17OHP)

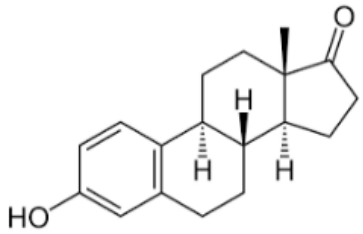
DHEA —3βHSD→ Androstenedione

17βHSD ↓ (DHEA) | 17βHSD ↓ (Androstenedione)

Androstenediol —3βHSD→ Testosterone

Standard Androgen Synthesis Pathway

Estrone
18 carbons



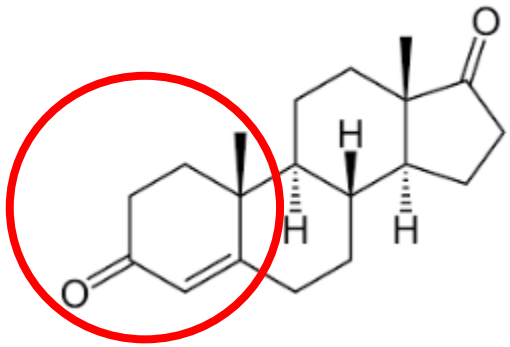
Aromatase: Androstenedione → Estrone

17βHSD: Estrone → Estradiol

Aromatase: Testosterone → Estradiol

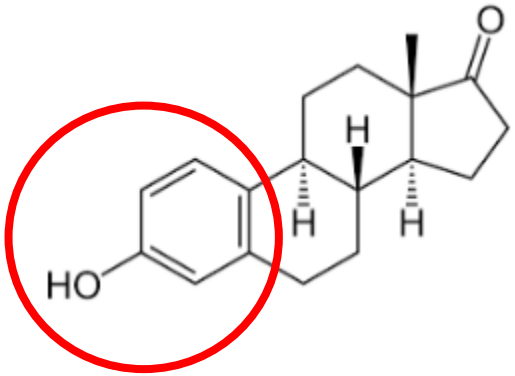
Estrogen Synthesis: **Aromatase**

Androstenedione
19 carbons



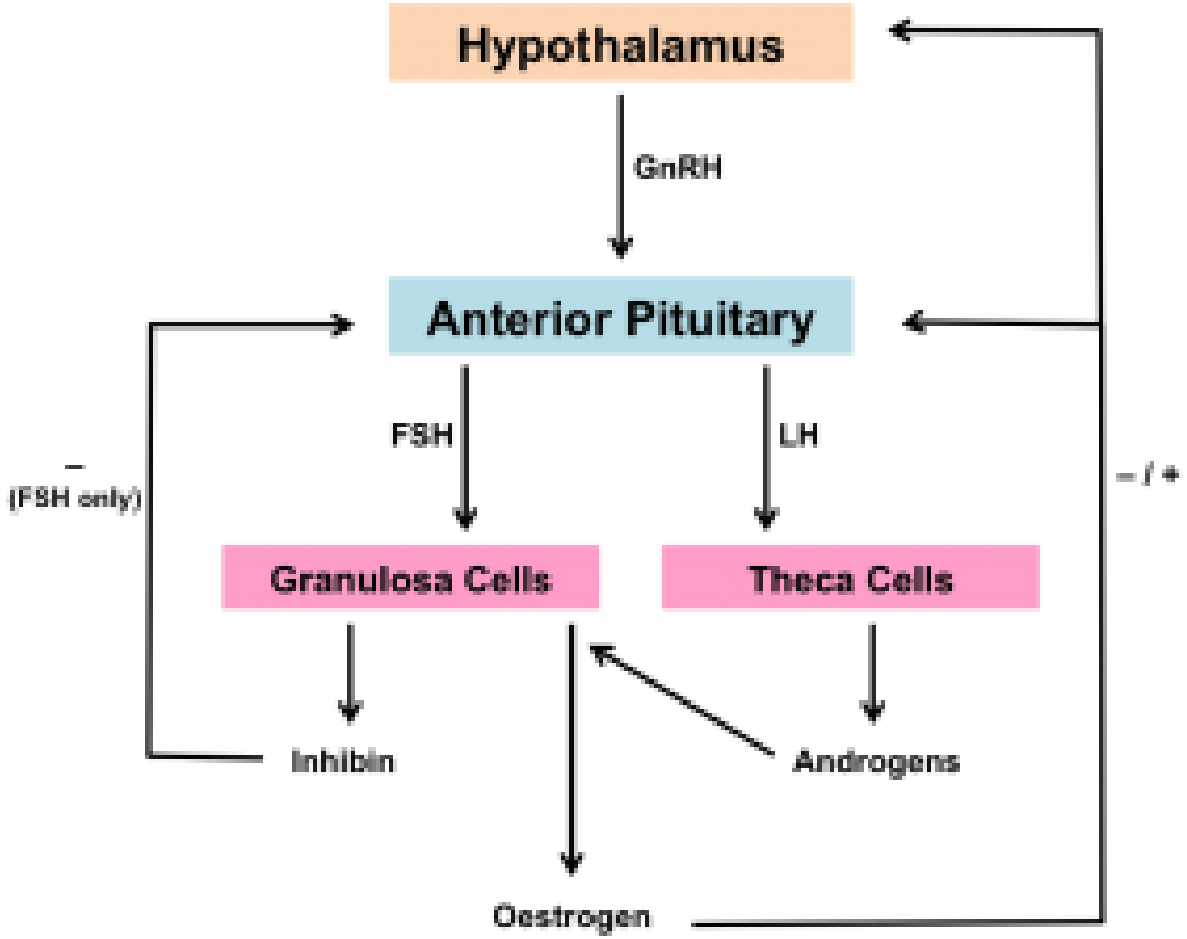
Aromatase

Estrone
18 carbons



- Expressed in steroidogenic tissue (ovaries, placenta), in brain, and in non-steroidogenic tissue (bone, fat)
- Aromatase in brain assists the neuroprotective effects of estrogen
- Aromatase in bone is critical to epiphyseal maturation
- Aromatase in placenta is critical in protecting mother from exposure to fetal androgens

Regulation of Estrogen Production: Hypothalamic-Pituitary-Ovary Axis



(Feedback regulation depends on phase of menstrual cycle)

GnRH Rec
GPCR (Gq/11)

LH Rec
GPCR (Gs)

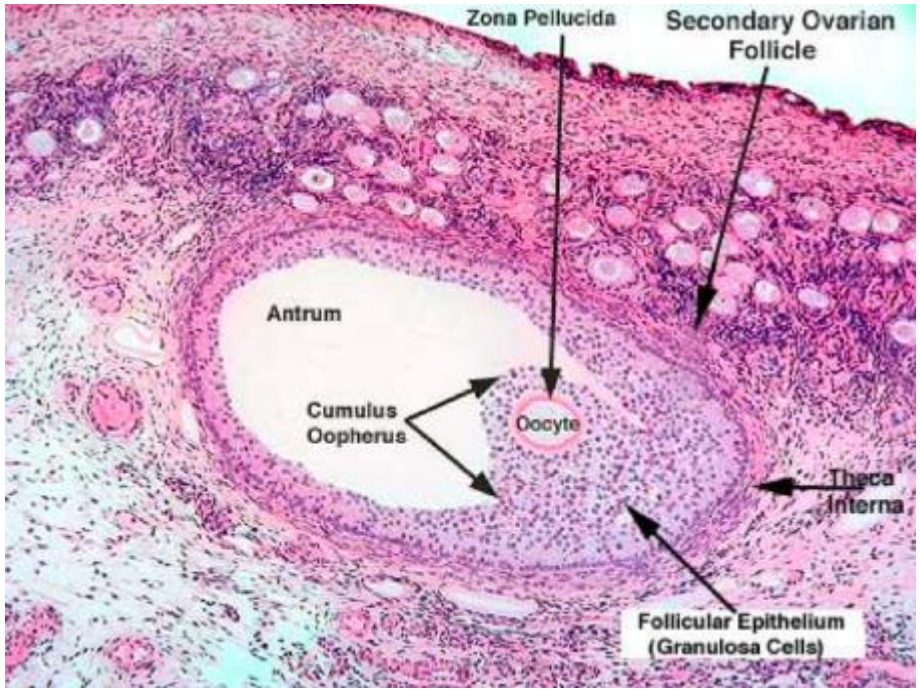
FSH Rec
GPCR (Gs)

Estrogen Rec
NHR

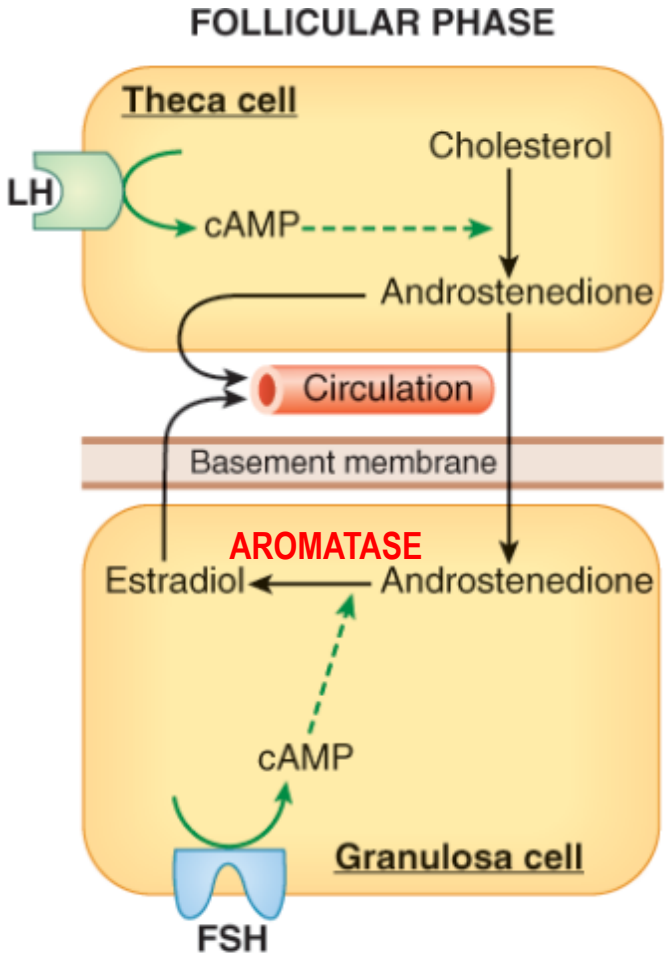
Inhibin blocks FSH production

Regulation of Ovarian Estrogen Production: Actions of LH and FSH in Follicular Phase

- LH and FSH receptors – Follicular Stage
  - G protein-coupled receptor coupled to Gs protein
  - Signaling cascade through cAMP and PKA
  - LH receptors expressed on **theca cells**
  - FSH receptors exclusively on **granulosa cells**

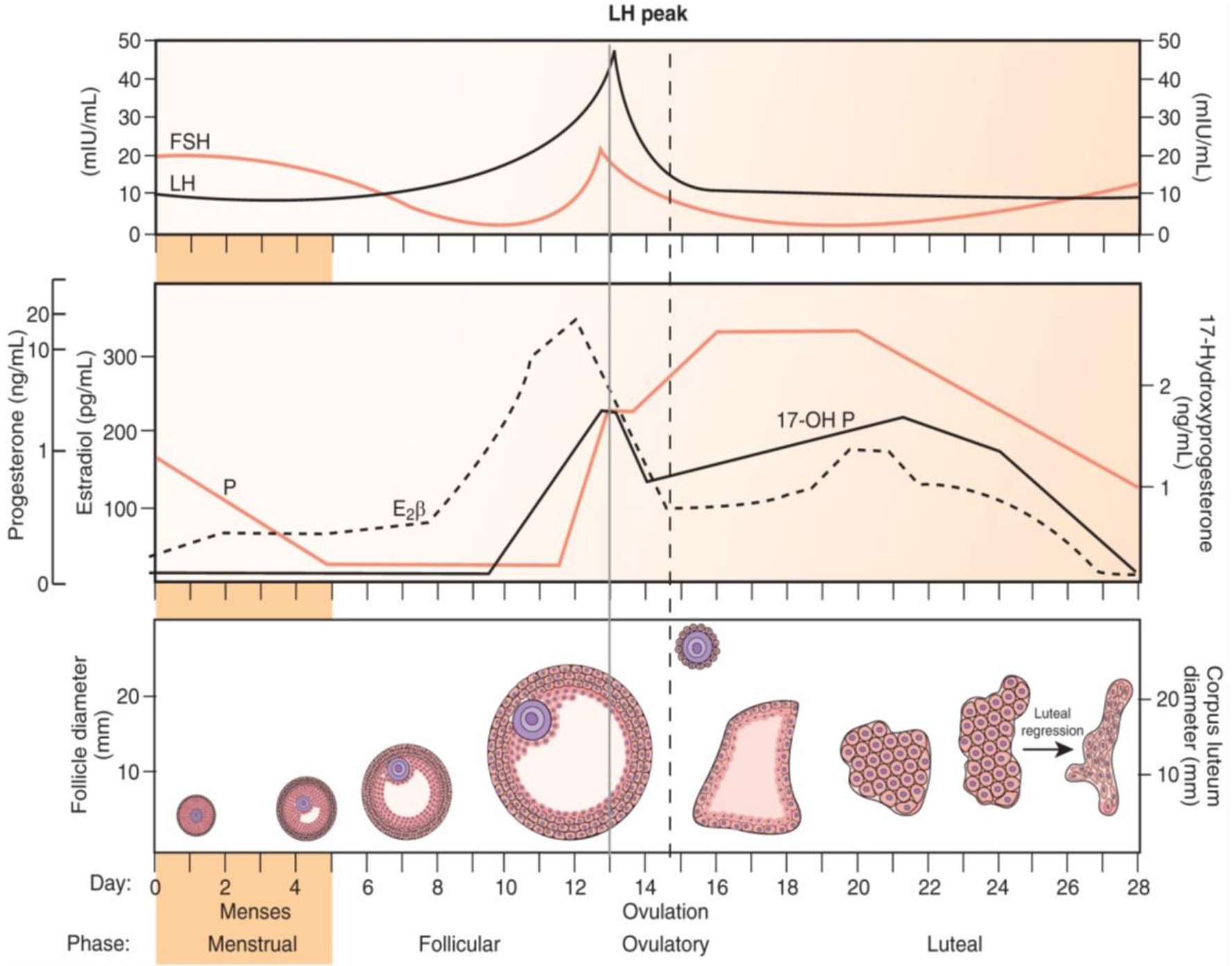


Regulation of Ovarian Estrogen Production: Actions of LH and FSH



In addition to stimulating aromatase activity in granulosa cells, FSH receptor activation also induces LH receptor expression.

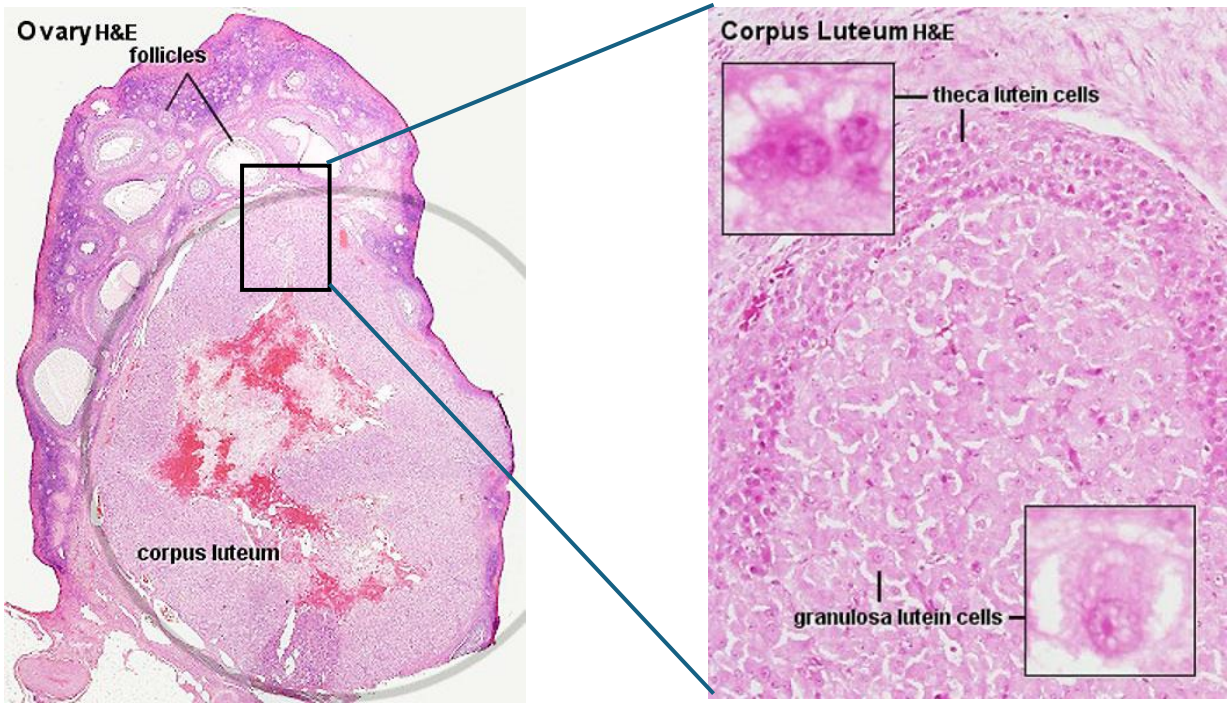
Source: F. Gary Cunningham, Kenneth J. Leveno, Steven L. Bloom, Catherine Y. Spong, Jodi S. Dashe, Barbara L. Hoffman, Brian M. Casey, Jeanne S. Sheffield: *Williams Obstetrics*, 25th Edition
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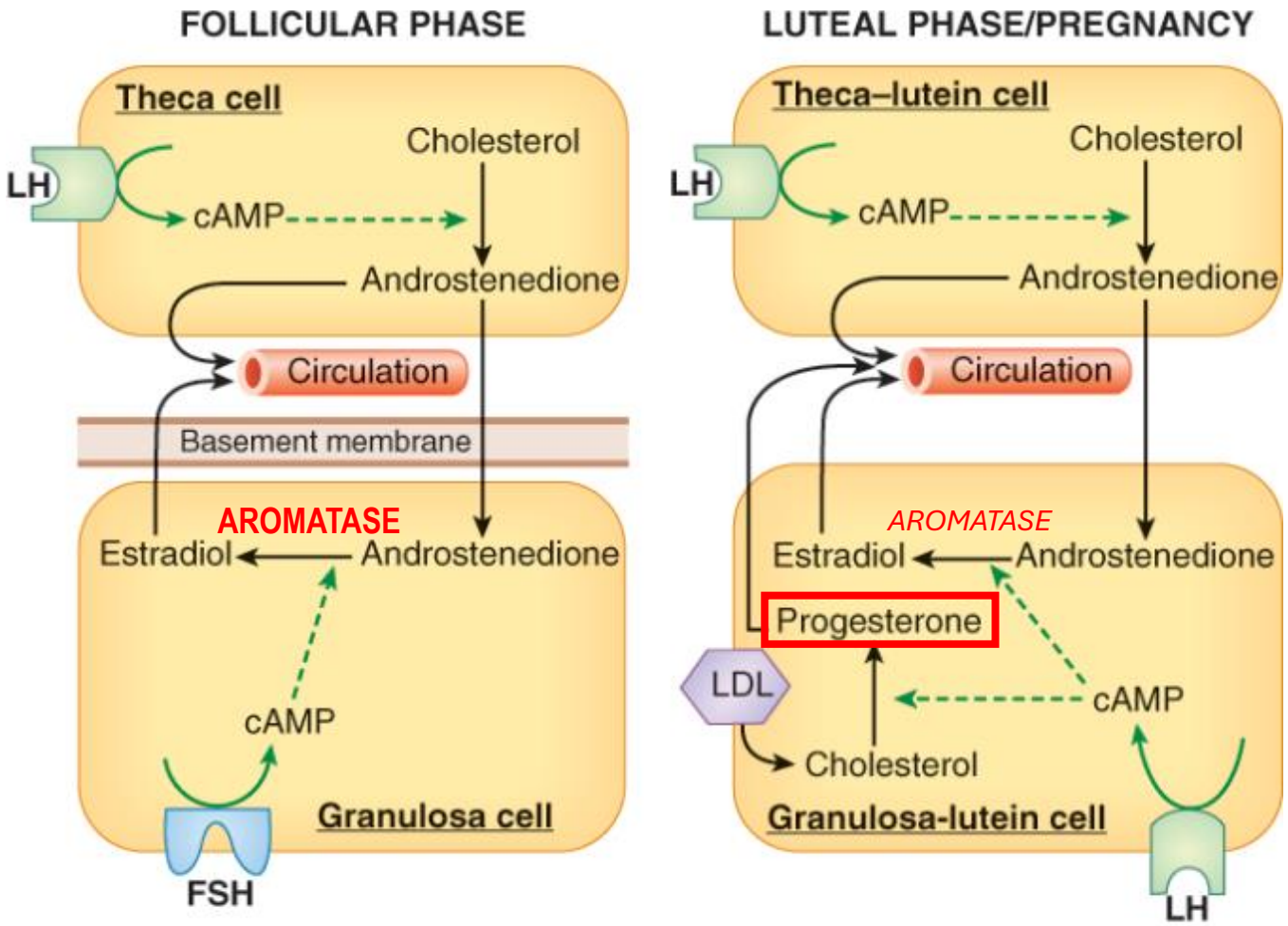
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Regulation of Ovarian Estrogen Production: Actions of LH during Luteal Phase

- LH and FSH receptors – Luteal phase
  - Gs signaling cascade through cAMP and PKA
  - LH receptors expressed on **theca-lutein cells** → stimulate conversion of cholesterol to adrostenedione
  - LH and FSH receptors expressed on **granulosa-lutein cells** → **stimulates conversion of cholesterol to progesterone** plus maintains conversion of androgens to estrogens



Regulation of Ovarian Estrogen Production: Actions of LH and FSH



Source: F. Gary Cunningham, Kenneth J. Leveno, Steven L. Bloom, Catherine Y. Spong, Jodi S. Dashe, Barbara L. Hoffman, Brian M. Casey, Jeanne S. Sheffield: *Williams Obstetrics*, 25th Edition
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Tissue-Specific Steroidogenic Pathways in the Ovaries

	Enzymes and Support Proteins Expressed	Major Steroid Produced
Theca cells (ovary)	Cholesterol SCC 3βHSD 17α-Hydroxylase Cytochrome b5 (17βHSD)	Androstenedione (Testosterone)
Granulosa cells (ovary)	**AROMATASE** **17βHSD** End of follicular phase→luteal phase Cholesterol SCC 3βHSD	(Estrone) Estradiol Progesterone

Learning Objectives

1. Map the pathways involved in synthesis of adrenal steroids and explain how tissue-specific production of corticosteroids is achieved.

2. Describe the hormonal regulation of adrenal steroid production and explain why chronic signaling leads to congenital adrenal hyperplasia (CAH).

3. Map the pathways involved in the synthesis of testosterone, estrogen and progesterone in gonadal tissue and describe their hormonal regulation.

4. Compare and contrast the effects on blood pressure, metabolism, stress response, and sexual development of common disorders of steroidogenesis (21-hydroxylase deficiency, 11β-hydroxylase deficiency, 17α-hydroxylase deficiency, 11β-hydroxysteroid dehydrogenase deficiency) and describe their pathophysiological mechanisms.

5. Outline common laboratory tests for identifying patients with defects in steroidogenesis and explain how they discriminate between 21-hydroxylase deficiency, 11β-hydroxylase deficiency and 17α-hydroxylase deficiency.

Case 2: …23-day old baby (*was*) referred to our hospital with complaints of discomfort and inability to gain weight. …external genital structure … phallus was 2.3 cm, bilateral scrotal structure was empty and gonads were unpalpable. Suitable water-electrolyte treatment was started after diagnosis of dehydration and surrenal (*adrenal*) deficiency with laboratory findings. In the abdominal ultrasonography … uterus and ovary structures were found and surrenal glands were bigger than normal size and in cerebriform structure.

Ozdemir, *et al., Ann Clin Case Rep*. (2018) **3**: 1544.

Question: What is responsible for virilization in this patient?

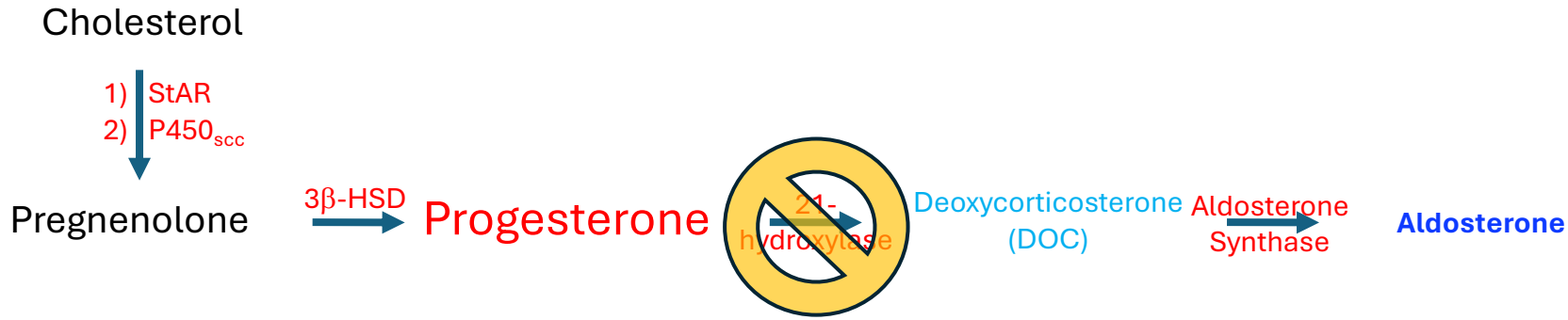
Disorders of Steroidogenesis

21-hydroxylase deficiency

- Incidence: 1 in 5000 to 1:15000 live births
- Most common cause of congenital adrenal hyperplasia (CAH)
- Caused by mutations in or deletion of 21-hydroxylase gene (Cyp21A2)
  - Disease requires homozygous or compound heterozygous mutations (both alleles must be affected)
- **Three forms** depending on residual enzymatic activity in steroidogenic tissue
  - Salt-wasting – endocrinological emergency diagnosed at birth
  - Simple virilizing – recognized at birth due to DSD
  - Cryptic or late-onset – often not diagnosed until childhood or early adulthood

Disorders of Steroidogenesis: 21-Hydroxyase

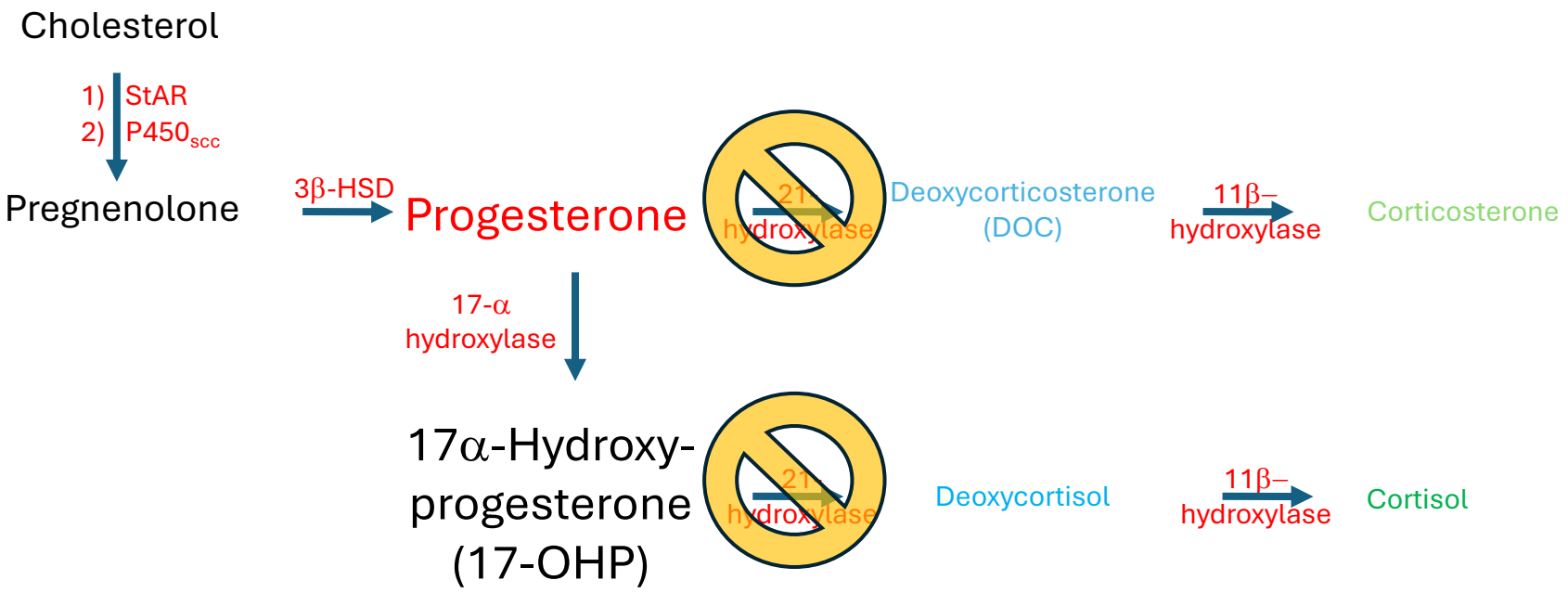
Zona Glomerulosa



- **21-hydroxylase deficiency**
  - **Impairs aldosterone production** from adrenal zona glomerulosa when residual enzymatic activity is less than 1% of normal
  - Salt Wasting form (enzymatic activity ~0% normal)
    - **Aldosterone ↓**
    - **Serum sodium ↓**
    - **Serum potassium ↑**
    - **Hypovolemia**
    - **Hypotension**
    - **Renin ↑, ATII ↑, CAH**

Disorders of Steroidogenesis: 21-Hydroxylase

Zona Fasciculata



- **21-hydroxylase deficiency**
  - **Impairs cortisol production** from adrenal zona fasciculata when residual enzymatic activity is less than 5% of normal
  - Salt Wasting and Simple Virilizing forms
    - **Serum cortisol ↓**
    - **ACTH ↑ CAH**

      Note that maternal cortisol cannot compensate for the lack of fetal cortisol due to 11βHSD in the placenta.

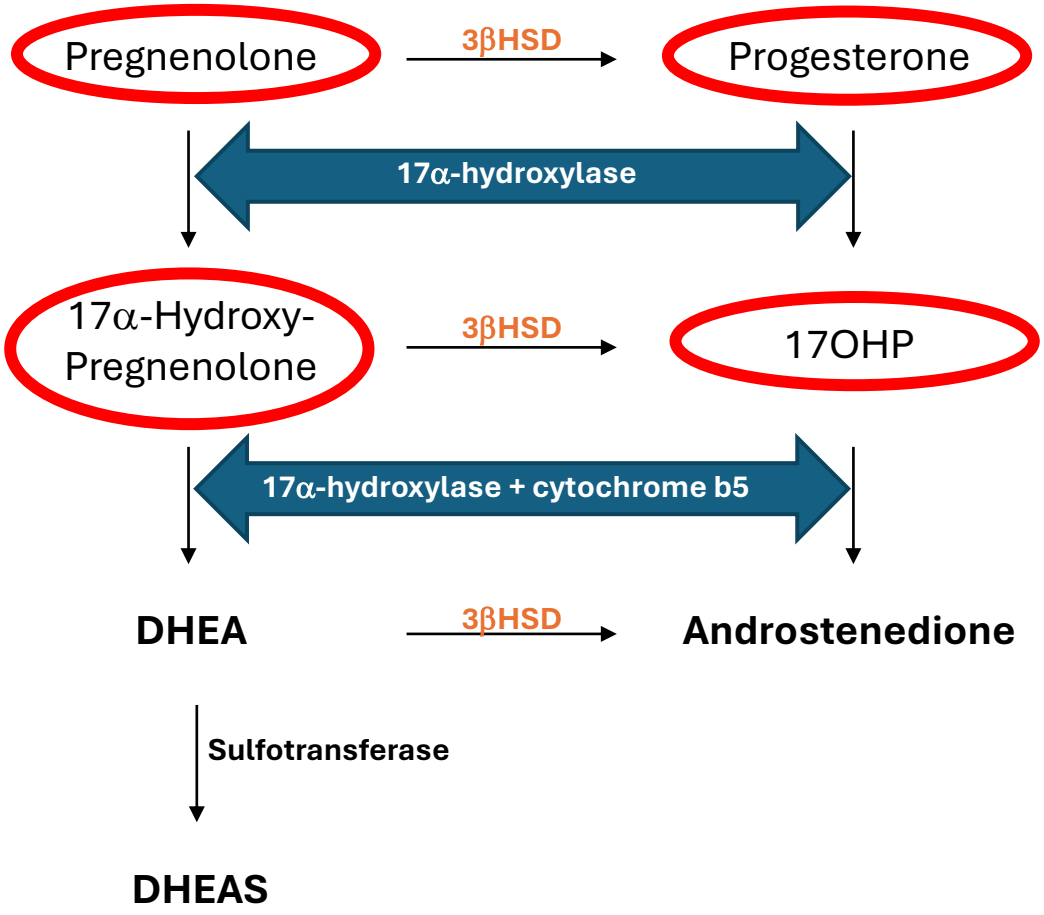
Case 2: …23-day old baby (*was*) referred to our hospital with complaints of discomfort and inability to gain weight. …external genital structure … phallus was 2.3 cm, bilateral scrotal structure was empty and gonads were un-palpable. Suitable water-electrolyte treatment was started after diagnosis of dehydration and surrenal (*adrenal*) deficiency with laboratory findings. In the abdominal ultrasonography … uterus and ovary structures were found and surrenal glands were bigger than normal size and in cerebriform structure.

Ozdemir, *et al., Ann Clin Case Rep*. (2018) **3**: 1544.

Disorders of Steroidogenesis: 21-Hydroxylase

21-hydroxylase deficiency

- Intermediates in failed cortisol synthesis (circled) diffuse to fetal layer to increase adrenal androgen synthesis
- High ACTH from low cortisol stimulates androgen synthesis in the fetal zone of the fetal adrenal
- High androgen levels cause virilization of genitalia in an XX fetus (referred to as “ambiguous genitalia” or “hermaphroditism” in older literature; now referred to as one type of difference of sexual development (DSD))



Case 2: …23-day old baby (*was*) referred to our hospital with complaints of discomfort and inability to gain weight. …external genital structure … phallus was 2.3 cm, bilateral scrotal structure was empty and gonads were un-palpable. Suitable water-electrolyte treatment was started after diagnosis of dehydration and surrenal (*adrenal*) deficiency with laboratory findings. In the abdominal ultrasonography … uterus and ovary structures were found and surrenal glands were bigger than normal size and in cerebriform structure.

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With karyotype and hormonal results, virilized female and salt depleting 21-hydroxylase deficiency were diagnosed. Treatment with hydrocortisone, mineralocorticoid and salt replacement was started. …deletion was detected in the CYP21A2 gene of the patient in line with 21-hydroxylase and salt wasting type.

Ozdemir, *et al., Ann Clin Case Rep*. (2018) **3**: 1544.

COMMONWEALTH OF PENNSYLVANIA • DEPARTMENT OF HEALTH
VITAL RECORDS

Certification of Birth

DATE OF BIRTH (MO DAY YEAR)

FILE NO

DATE FILED (MO DAY YEAR)

COUNTY OF BIRTH

DATE ISSUED (MO DAY YEAR)

NAME

SEX

FATHERS NAME

MOTHERS MAIDEN NAME

This is to certify that this is a true copy of the record which is on file in the Pennsylvania Department of Health, in accordance with Act 66, P.L. 304, approved by the General Assembly, June 29, 1953.

Charles Hardester

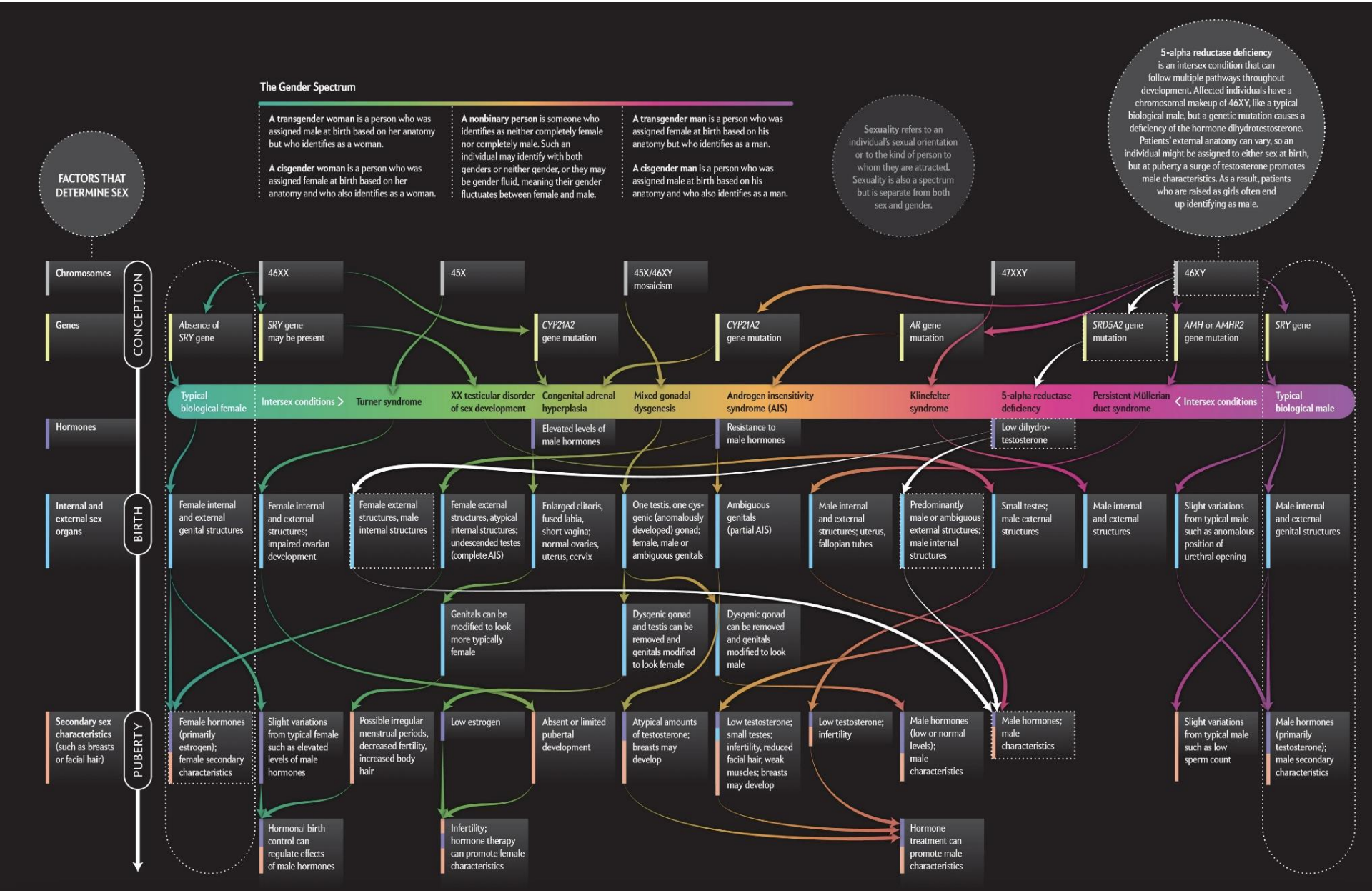
CHARLES HARDESTER
STATE REGISTRAR

H105.100 (REV. 01-99)

COMMONWEALTH OF PENNSYLVANIA
DEPARTMENT OF HEALTH

WARNING: IT IS ILLEGAL TO DUPLICATE THIS COPY BY PHOTOSTAT OR PHOTOGRAPH

2017



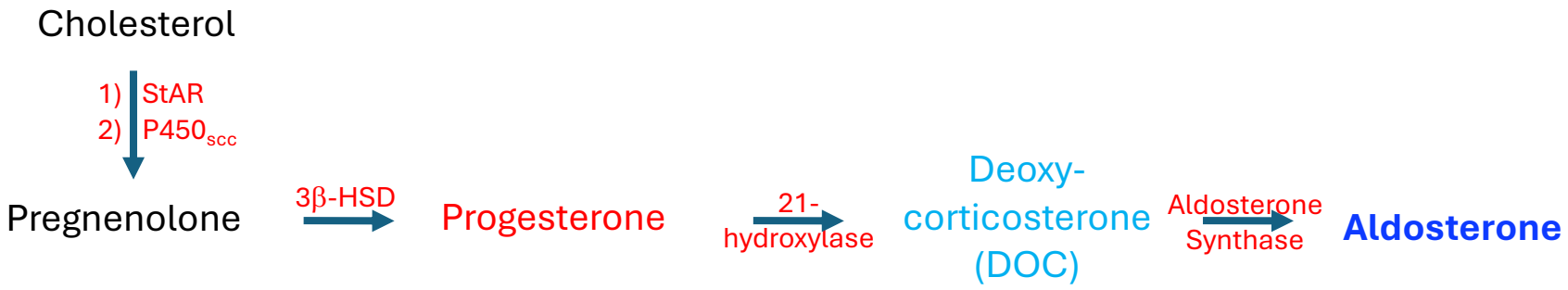
https://www.scientificamerican.com/article/beyond-xx-and-xy-the-extraordinary-complexity-of-sex-determination/

Disorders of Steroidogenesis

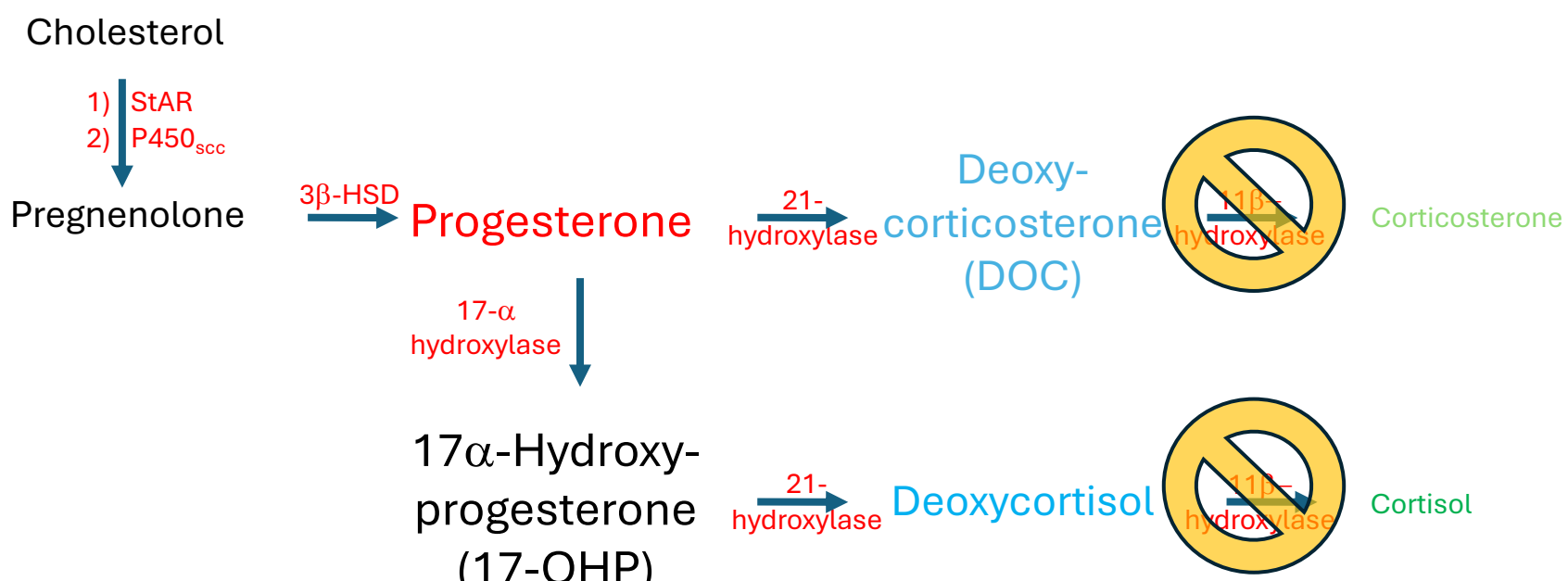
- **11β-hydroxylase deficiency**
  - Incidence: 1:100,000 live births
  - **Accounts for 7% of CAH cases**
  - Caused by mutations in or deletion of 11β-hydroxylase gene
    - Disease requires homozygous or compound heterozygous mutations (both alleles must be affected)
  - **Defects**
    - Cortisol production is severely decreased
    - Mineralocorticoid intermediates (e.g., DOC) of the cortisol synthesis pathway are produced in excess, leading to hypertension, low renin secretion and low aldosterone production.

Disorders of Steroidogenesis: 11β-Hydroxylase

Zona Glomerulosa

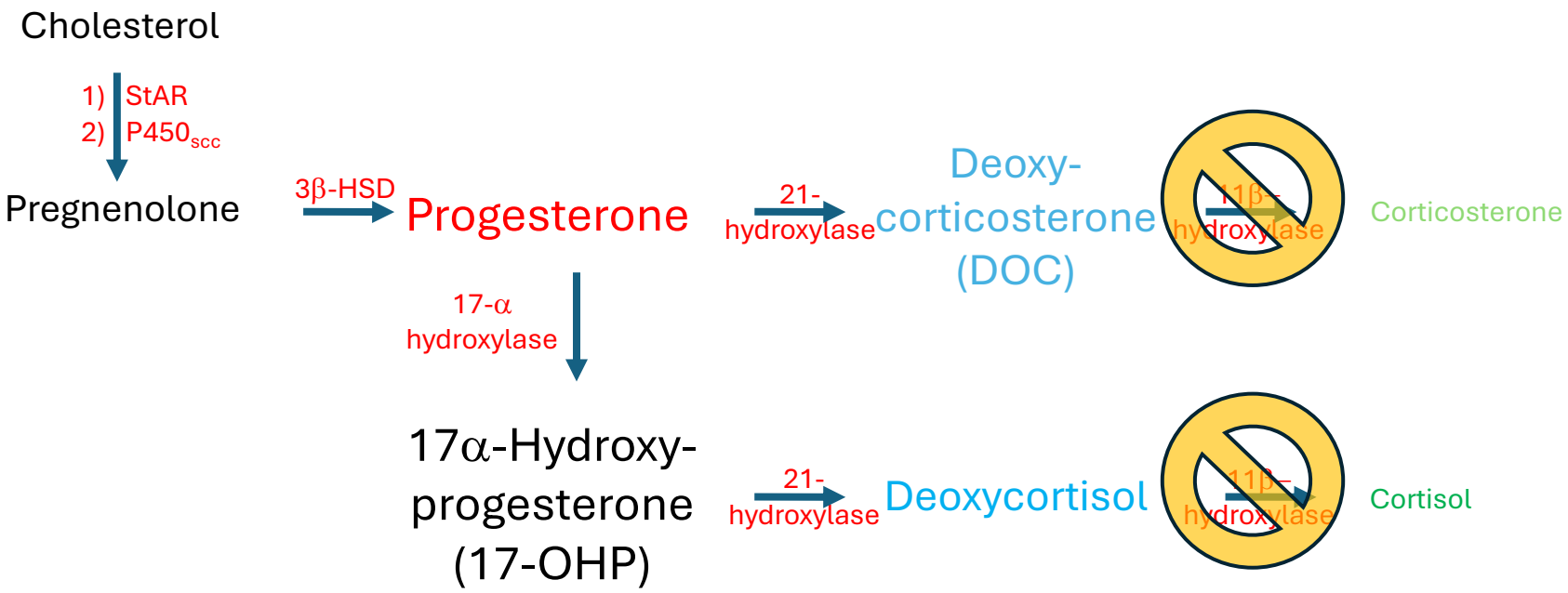


Zona Fasciculata



Disorders of Steroidogenesis: 11β-Hydroxylase

Zona Fasciculata



- **11β-hydroxylase deficiency**
  - Cortisol ↓ → ACTH ↑ → **CAH**
  - Mineralocorticoids DOC/Deoxycortisol ↑
  - Serum sodium ↑ → hypertension → renin ↓ → aldosterone ↓
  - Serum potassium ↓

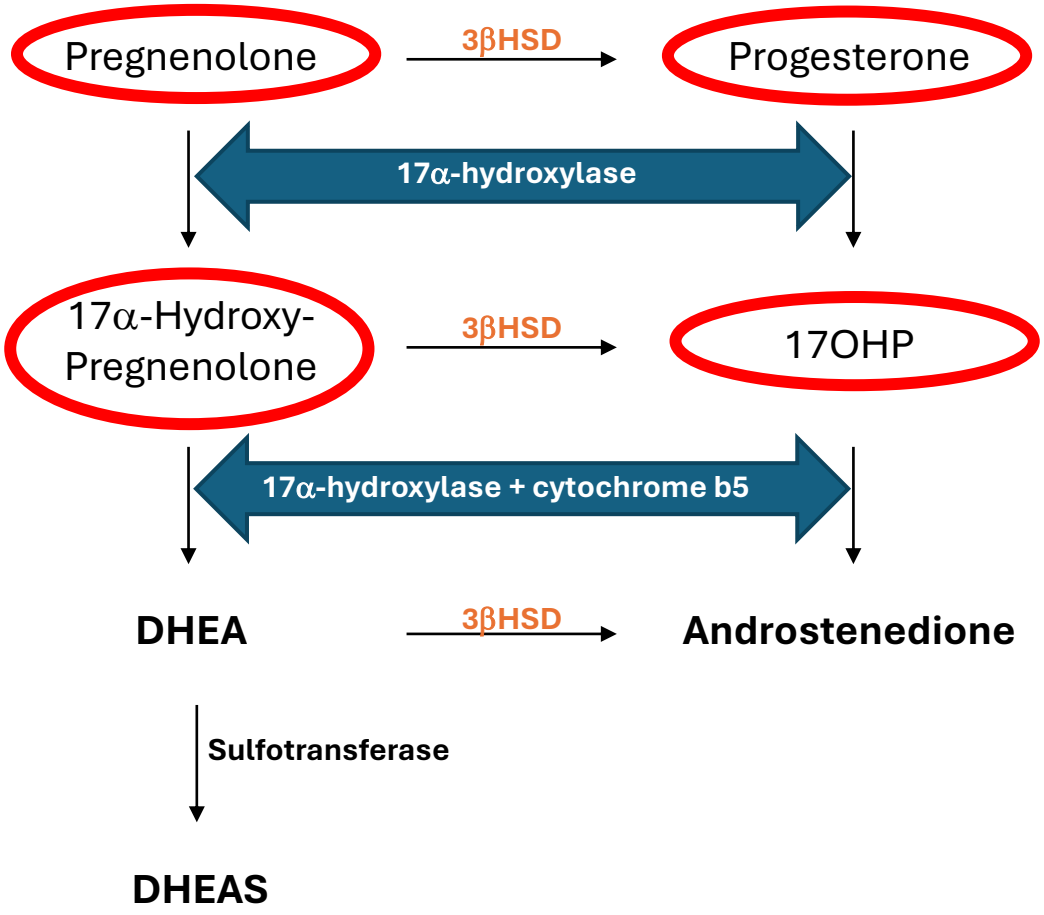
Disorders of Steroidogenesis: 11β-Hydroxylase

- **11β-hydroxylase deficiency**

  - Cortisol deficiency leads to inappropriately high ACTH and, consequently, CAH
  - Mineralocorticoid intermediates of cortisol synthesis
    DOC/Deoxycortisol are elevated, leading to:
    - Hypernatremia
    - Hypokalemia
    - Hypertension
    - Low renin → low aldosterone

  - Adrenal androgen production is enhanced due to
    - High ACTH increasing Zona reticularis cell mass
    - High levels of steroid precursors from failed cortisol synthesis in zona fasciculata

VIRILIZATION



Case 3: A 40-year-old…patient presented with…lack of breast development, complete absence of axillary and pubic hair, tall stature, and slenderness. The external genitalia were phenotypically female, the height was 175 cm, the weight was 64 Kg, and the body mass index (BMI) was 20.9. On admission, the blood pressure (BP) was persistently high throughout the day: mean systolic BP values were 170 mmHg; mean diastolic BP values were 110 mmHg. …biochemical and endocrine lab tests…showed the following: low-undetectable levels of testosterone, DHEA-S, androstenedione, 17-OH-progesterone, estradiol and active renin; low-normal 24-h urinary cortisol and serum aldosterone; high ACTH, progesterone, LH and FSH concentrations. Chromosomal analysis demonstrated a…

Papi, *et al., Case Rep. Endo.* (2018);
https://doi.org/10.1155/2018/2086861

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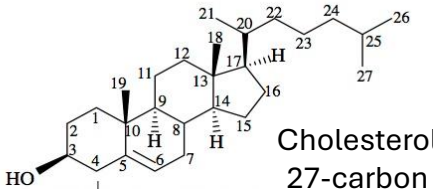
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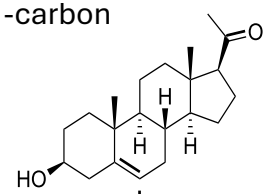
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Disorders of Steroidogenesis

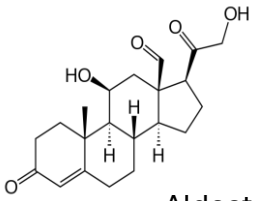


Cholesterol
27-carbon

Pregnenolone
21-carbon



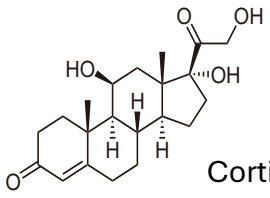
3βHSD → Progesterone → 21-Hydroxylase → **Aldosterone Synthase**



Aldosterone
21-carbon

17α-Hydroxylase

3βHSD → 17-OHP → 21-Hydroxylase → **11β-hydroxylase**

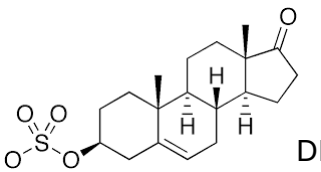


Cortisol
21-carbon

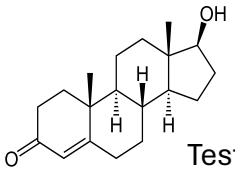
17α-Hydroxylase +
Cytochrome b5

3βHSD → 17βHSD

Sulfotransferase

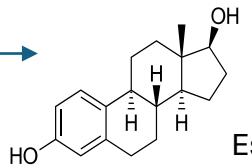


DHEAS
19-carbon



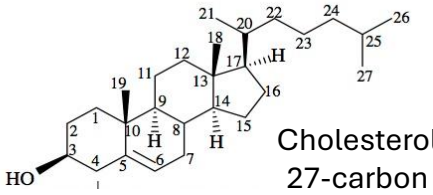
Testosterone
19-carbon

Aromatase



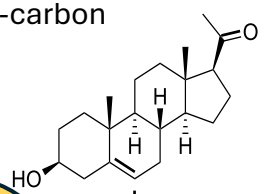
Estradiol
18-carbon

Disorders of Steroidogenesis

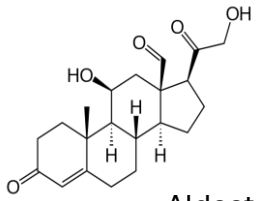


Cholesterol
27-carbon

Pregnenolone
21-carbon



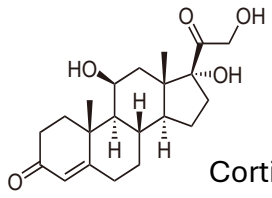
3βHSD → Progesterone → 21-Hydroxylase → **Aldosterone Synthase**



Aldosterone
21-carbon

17α-Hydroxylase

3βHSD → 17-OHP → 21-Hydroxylase → **11β-hydroxylase**

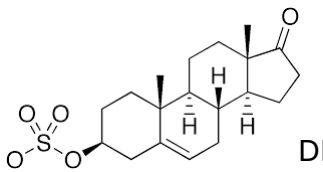


Cortisol
21-carbon

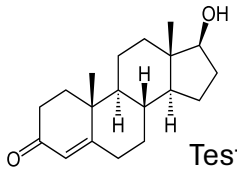
17α-Hydroxylase + Cytochrome b5

Sulfotransferase

3βHSD → 17βHSD

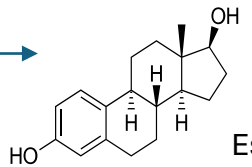


DHEAS
19-carbon



Testosterone
19-carbon

Aromatose



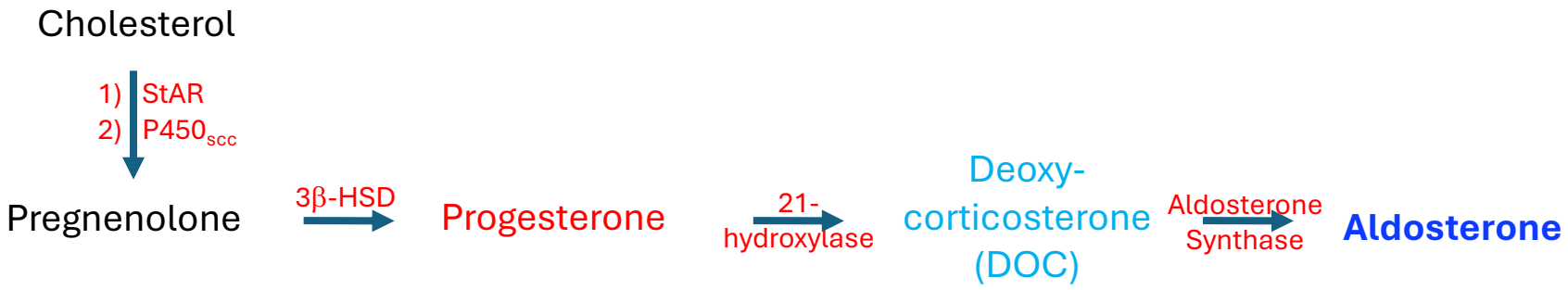
Estradiol
18-carbon

Disorders of Steroidogenesis

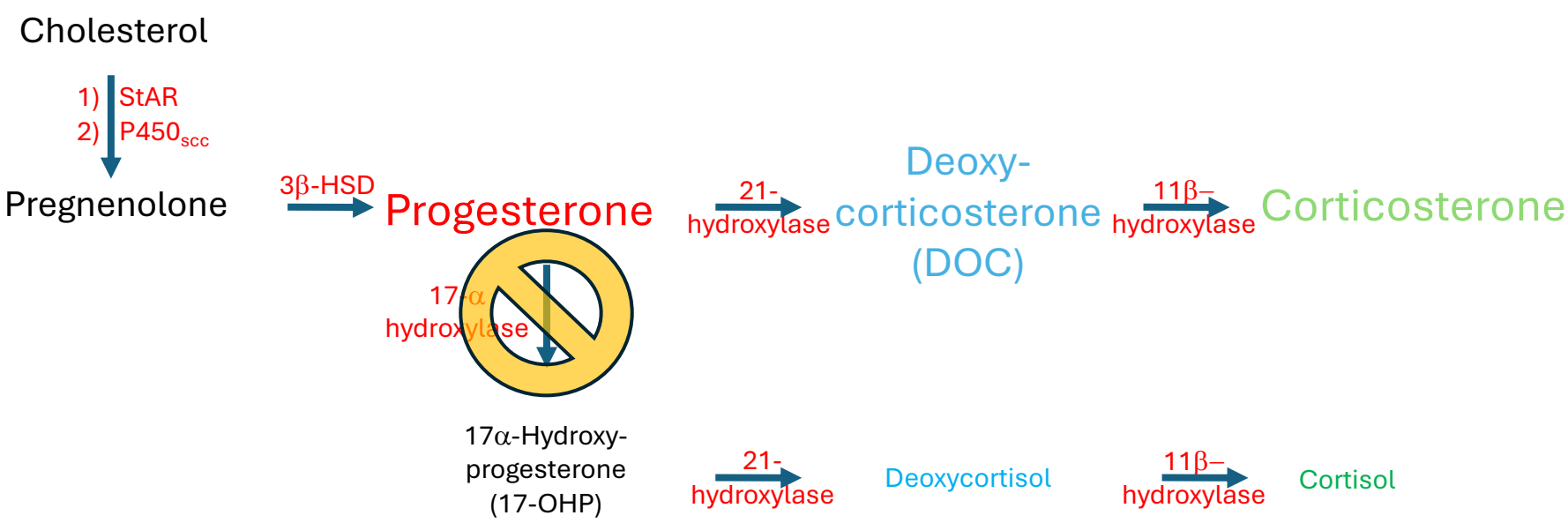
- **17α-hydroxylase deficiency**
  - Incidence:
    - **Accounts for < 1% of CAH cases**
    - Incomplete loss of enzymatic activity *may account for a significant percentage of primary hypertension* (based on GWAS analysis)
  - Caused by mutations in or deletion of 17α-hydroxylase gene
    - Disease requires homozygous or compound heterozygous mutations (both alleles must be affected)
  - **Defects**
    - Cortisol production
    - Aldosterone production (renin release is suppressed by mineralocorticoid activity of DOC)
    - Androgen and estrogen production

Disorders of Steroidogenesis: 17α-Hydroxylase

Zona Glomerulosa

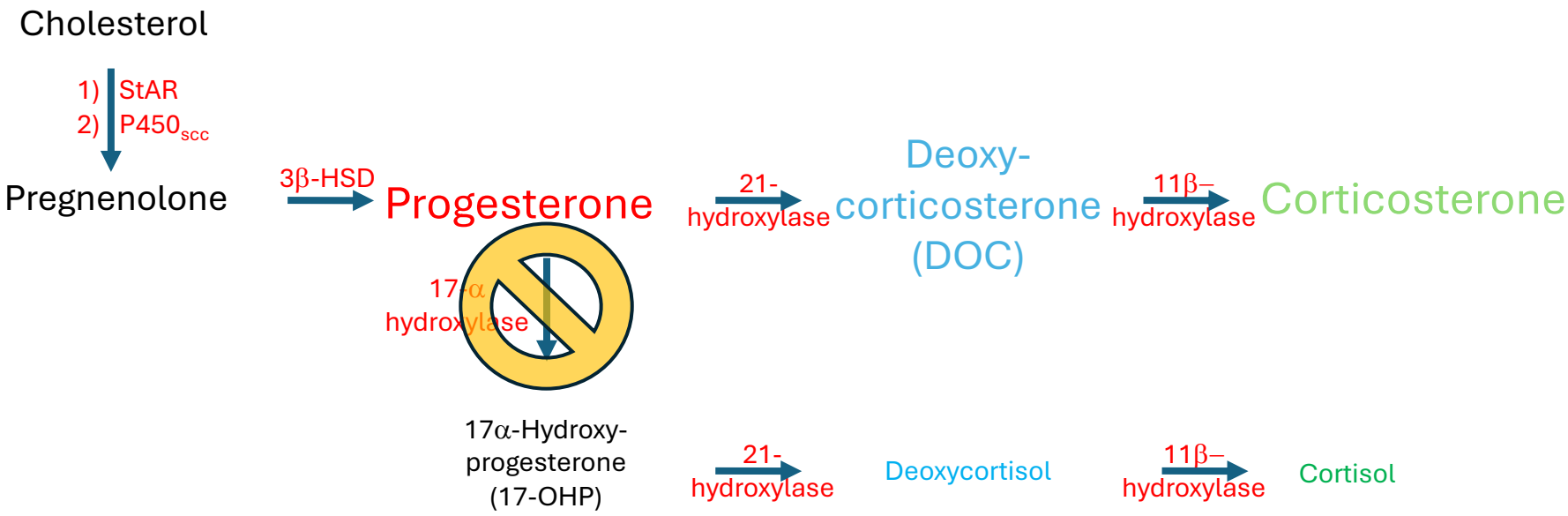


Zona Fasciculata



Disorders of Steroidogenesis: 17α-Hydroxylase

Zona Fasciculata



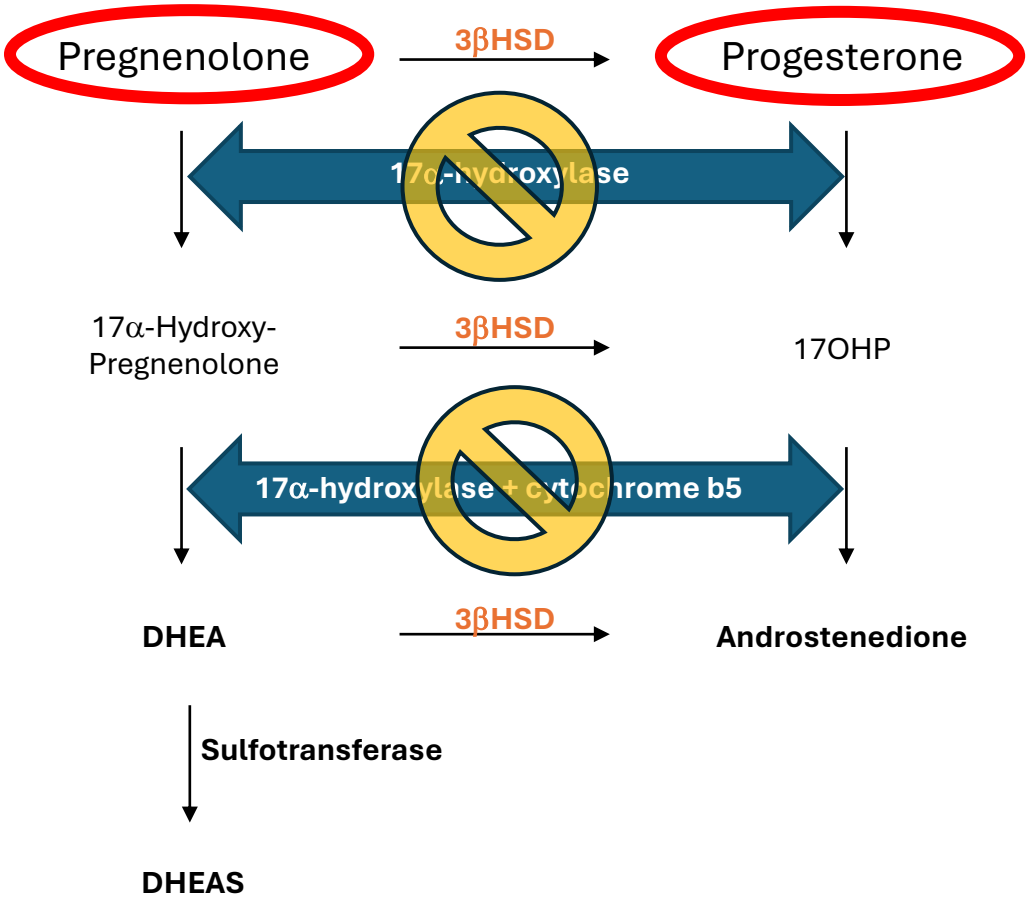
- **17α-hydroxylase deficiency**
  - Cortisol ↓ → ACTH ↑ → **CAH**
  - Mineralocorticoids DOC ↑
  - Serum sodium ↑ → hypertension → renin ↓ → aldosterone ↓
  - Serum potassium ↓

Disorders of Steroidogenesis: 17α-Hydroxylase

- **17α-hydroxylase deficiency**

  - Cortisol deficiency leads to inappropriately high ACTH and, consequently, CAH
  - Mineralocorticoid intermediate of cortisol synthesis DOC is elevated, leading to:
    - Hypernatremia
    - Hypokalemia
    - Hypertension
    - Low renin → low aldosterone

  - Adrenal androgen production is compromised despite high ACTH increasing Zona reticularis cell mass and high levels of progesterone from failed cortisol synthesis in zona fasciculata
  - Gondal androgen and estrogen production is compromised.



NON-VIRILIZING

- XX infants appear normal; problems arise in puberty
- XY infants show absent or incomplete sexual development

Learning Objectives

1. Map the pathways involved in synthesis of adrenal steroids and explain how tissue-specific production of corticosteroids is achieved.

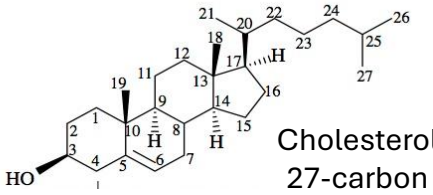
2. Describe the hormonal regulation of adrenal steroid production and explain why chronic signaling leads to congenital adrenal hyperplasia (CAH).

3. Map the pathways involved in the synthesis of testosterone, estrogen and progesterone in gonadal tissue and describe their hormonal regulation.

4. Compare and contrast the effects on blood pressure, metabolism, stress response, and sexual development of common disorders of steroidogenesis (21-hydroxylase deficiency, 11β-hydroxylase deficiency, 17α-hydroxylase deficiency, 11β-hydroxysteroid dehydrogenase deficiency) and describe their pathophysiological mechanisms.

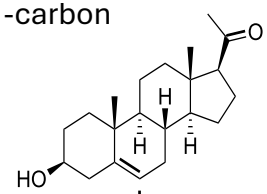
5. Outline common laboratory tests for identifying patients with defects in steroidogenesis and explain how they discriminate between 21-hydroxylase deficiency, 11β-hydroxylase deficiency and 17α-hydroxylase deficiency.

Newborn Screening for CAH

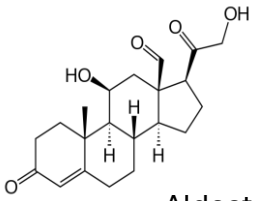


Cholesterol
27-carbon

Pregnenolone
21-carbon



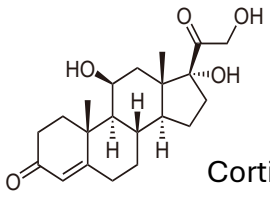
3βHSD → Progesterone → 21-Hydroxylase → **Aldosterone Synthase**



Aldosterone
21-carbon

17α-Hydroxylase

3βHSD → 17-OHP → 21-Hydroxylase → **11β-hydroxylase**



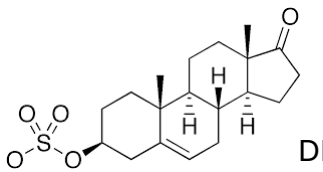
Cortisol
21-carbon

17α-Hydroxylase +
Cytochrome b5

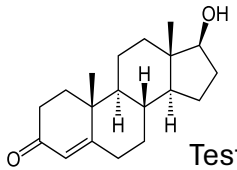
3βHSD

Sulfotransferase

17βHSD

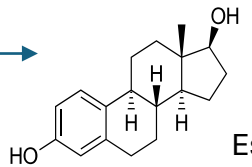


DHEAS
19-carbon



Testosterone
19-carbon

Aromatose



Estradiol
18-carbon

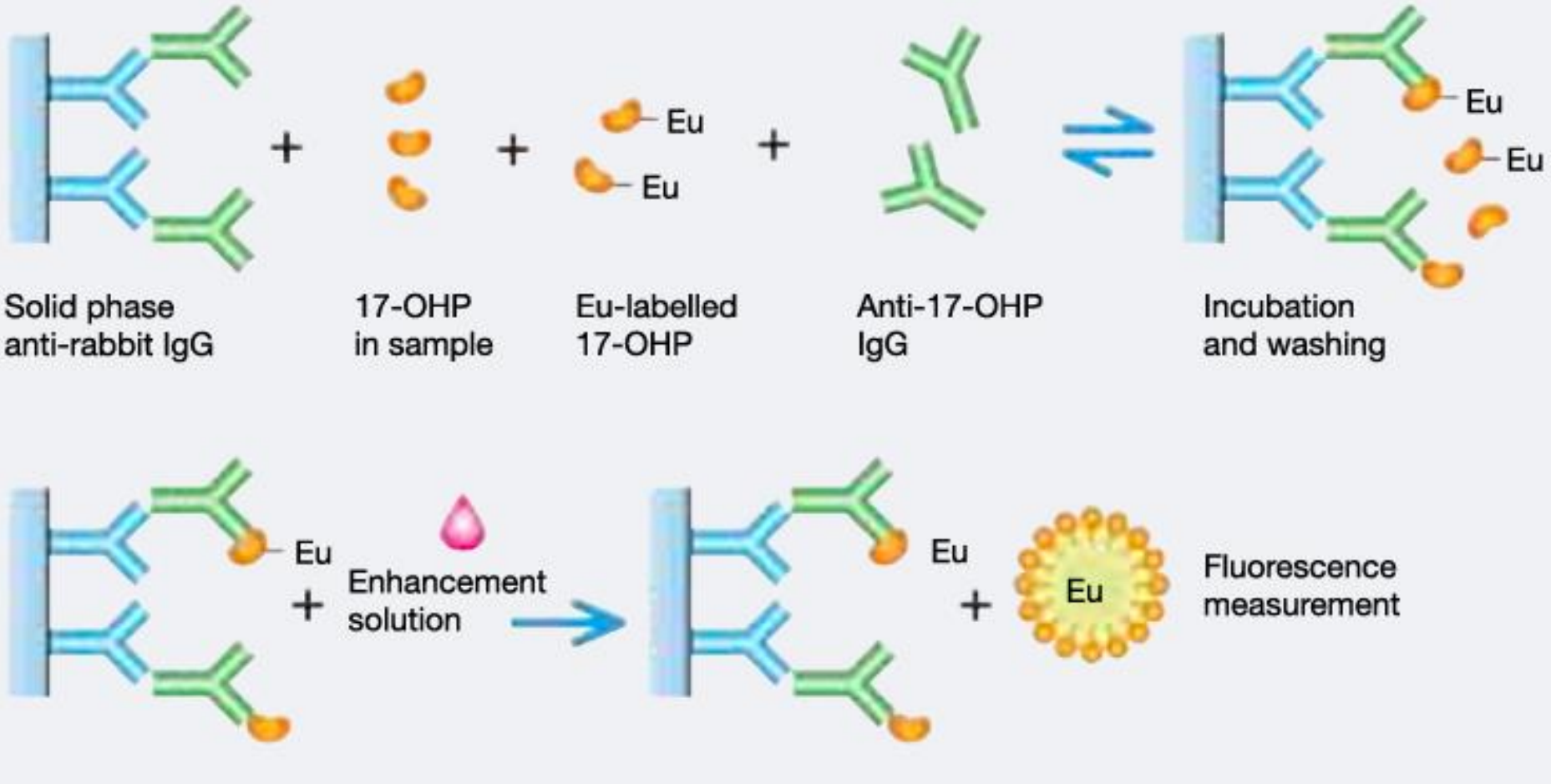
Newborn Screening for CAH



Photo courtesy of Dr. Brad Therrell

https://www.cdc.gov/nbslabbulletin/bulletin_laboratory_role.html

Competitive ELISA assay for 17-OHP



resources.perkinelmer.com

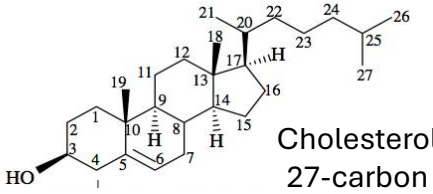
ACTH Stimulation Test

Used to diagnose late onset/cryptic congenital adrenal hyperplasia and adrenal insufficiency (Addison's disease)

Protocol:

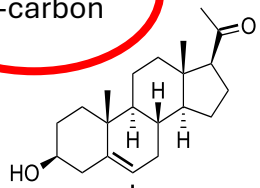
1) Peripheral blood drawn to establish patient baseline for corticosteroid analytes (e.g., pregnenolone,17-OHP, cortisol, DHEA).
2) Patient is stimulated with synthetic ACTH (e.g., 250µg, IV or IP).
3) Peripheral blood is drawn one hour post stimulation to determine effect of ACTH on analyte levels.

Newborn Screening for CAH



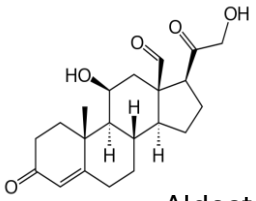
Cholesterol
27-carbon

Pregnenolone
21-carbon



3βHSD 21-Hydroxylase **Aldosterone Synthase**

Progesterone

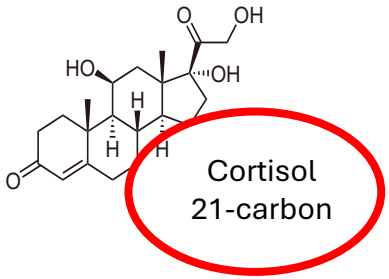


Aldosterone
21-carbon

17α-Hydroxylase

3βHSD 21-Hydroxylase **11β-hydroxylase**

17-OHP



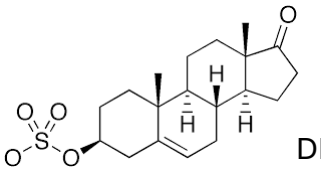
Cortisol
21-carbon

17α-Hydroxylase +
Cytochrome b5

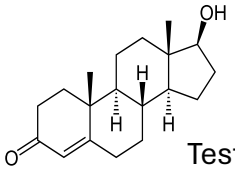
DHEA 3βHSD

Sulfotransferase

17βHSD

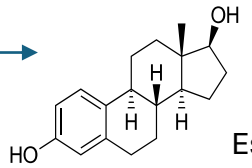


DHEAS
19-carbon



Testosterone
19-carbon

Aromatose



Estradiol
18-carbon

Summary: clinical presentation of steroidogenesis defects

Disorder	Complaint	DSD	Newborn Screen	ACTH challenge
21OHD (> 90%)	**Hypotension**	**Virilizing (in XX)**	**High 17-OHP**	Preg + **17-OHP +++** **Cortisol –** **DHEA +++**
11βOHD (< 10%)	**Hypertension**	**Virilizing (in XX)**	**High 17-OHP**	Preg + **17-OHP ++** **Cortisol –** **DHEA +++**
17αOHD (< 1%)	**Hypertension**	**Non-virilizing* (XY with female characteristics)**	**Low (normal) 17-OHP**	Preg + **17-OHP –** **Cortisol –** **DHEA –**

Case 5: A tall, slender, 18-year-old patient was referred to the cardiology department for persistent hypertension (since age 15), with a maximal reading of 180/110 mm Hg. The patient was treated with an angiotension II receptor antagonist (telmisartan) and a calcium channel blocker (amlodipine). However, her blood pressure was still not well controlled.

(Clin Exp Reprod Med. 2015 Jun; 42(2): 72–76.
doi: 10.5653/cerm.2015.42.2.72)

Question: What is the differential diagnosis for persistent hypertension in an adolescent?

1) Primary hypertension
2) Secondary hypertension
    a) Renal parenchymal disease
    b) Vascular disease
    c) Endocrine disorder

Case 2: ACTH stimulation test showed no increase in 17-OHP after intravenous administration of synthetic ACTH, suggesting an impairment in 17α-hydroxylase. Genetic analysis showed a compound heterozygous mutation in exon 6 of the *CYP17A1* gene (c.985-987TAC>AA and c.1118A>T).

(Clin Exp Reprod Med. 2015 Jun; 42(2): 72–76.
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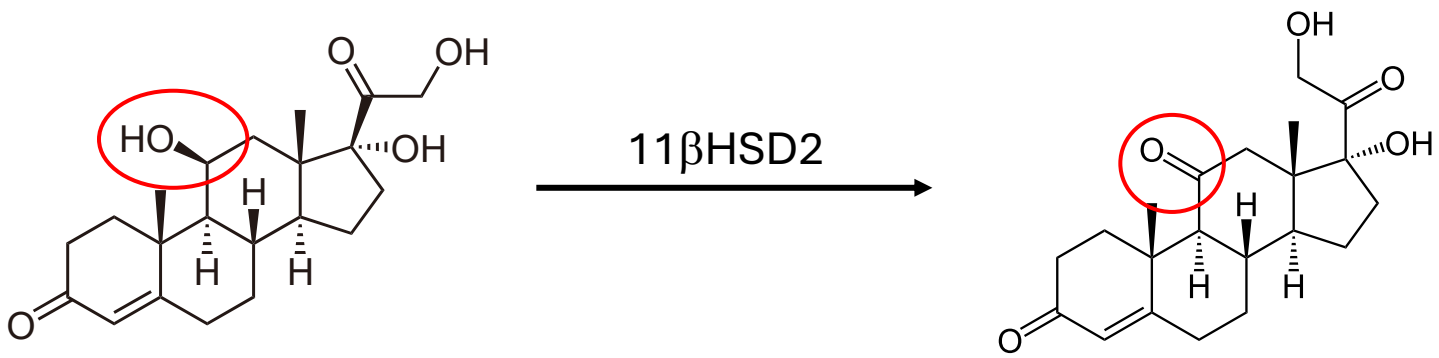
After being diagnosed with 17α-hydroxylase deficiency, the patient stopped taking antihypertensive medication and **started oral hydrocortisone**. The patient's blood pressure normalized without antihypertensive therapy, and follow up laboratory investigations revealed normal levels of ACTH (11.04 μg/dL), progesterone (0.04 ng/mL) and serum electrolytes.

(Clin Exp Reprod Med. 2015 Jun; 42(2): 72–76.
doi: 10.5653/cerm.2015.42.2.72)

Disorders of Steroid Modification

Tissue-Specific Steroid <u>Modification</u> in Aldosterone Signaling

- <u>Problem:</u> Aldosterone receptor (NHR) binds equally well to aldosterone and cortisol. But cortisol levels are 1000-fold higher than aldosterone levels.
- <u>Solution:</u> To sense aldosterone levels, aldosterone responsive cells must first inactivate cortisol.
- 11βHSD (type 2) catalyzes oxidation of cortisol into cortisone
  - Found in distal nephron (also placenta and colon)

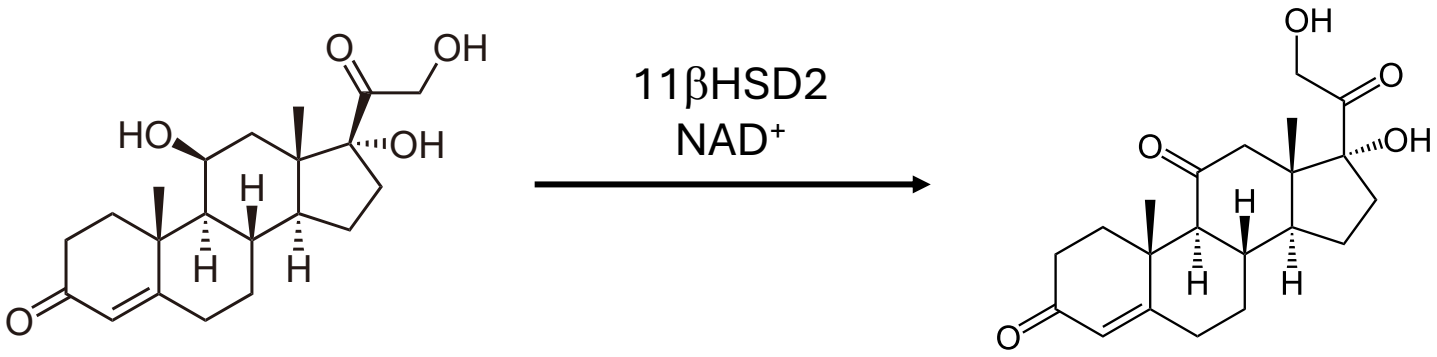


Cortisol
(Active)

Cortisone
(Inactive)

Disorders of Steroid Modification

- Apparent mineralocorticoid excess (AME)
  - Caused by mutations in 11βHSD2
  - Symptoms: hypervolemic HTN, salt retention, hypokalemia
  - Labs: no mineralocorticoids, very low renin
  - Very serious condition - can lead to end-organ failure at a young age if untreated.
  - Treat with aldosterone receptor antagonists, low salt diet. Even with treatment, 10% die of CVA



Cortisol
(Active)

Cortisone
(Inactive)

Summary

1. In XY males, *de novo* testosterone production follows the synthesis pathway of adrenal androgens but is restricted to the testes (and ovaries) through tissue-specific expression of **17βHSD**. Steroidogenesis in the **testes** is stimulated by **LH**. Testosterone DOES feed back on LH (and GnRH).
2. Testosterone affinity for the androgen receptor is enhanced by reduction to dihydrotestosterone (DHT) in target tissues through tissue-specific expression of **5α-reductase.** DHT is necessary for full receptor agonism and correct sexual development.
3. Ovarian **theca cells are** stimulated by **LH** to produce androstenedione (end testosterone). Ovarian granulosa cells stimulated by FSH convert androgens to estradiol and estrone via aromatase. During luteal phase, LH and FSH stimulation of granulosa-lutein cells promotes progesterone synthesis.
4. In CAH caused by enzyme deficiencies, hypernatremia and hypertension is observed due to the excess of steroid intermediates with mineralocorticoid activity (**11β-hydroxylase and 17α-hydroxylase).** In **21-hydroxylase deficiency,** there are no such intermediates synthesized, resulting in hyponatremia and hypotension.
5. In CAH**, virilization** is the result of enhanced androgen production due to unregulated ACTH stimulation of the zona reticularis and presence of presence of androgen precursors from failed synthesis pathways (**21-hydroxylase and 11β-hydroxylase**). In **17α-hydroxylase** deficiency, adrogen production is suppressed and virilization is not observed.
6. Newborn 17OHP testing identifies CAH caused by 21OHD & 11βOHD, but not 17αOHD. Diagnosis of 17αOHD requires ACTH challenge test.